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REVIEW

Monocyte-mediated mechanisms in idiopathic pulmonary fibrosis: opportunities for early intervention

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Abstract
Idiopathic pulmonary fibrosis (IPF) is a chronic interstitial lung disease marked by irreversible deposition of the extracellular matrix (ECM) and subsequent disruption of pulmonary architecture. Although current pharmacological interventions, such as Pirfenidone and Nintedanib, are available, they merely decelerate the progression of the disease. Notably, the monocyte count in peripheral blood is strongly correlated with the prognosis and mortality associated with IPF. An elevated monocyte count is observable in the early stages of IPF, with monocyte accumulation in lung tissue persisting throughout the disease's progression. Monocytes are recruited to the lung tissue in response to chemoattractant signals, where they differentiate into macrophages, dendritic cells, and fibrocytes. These differentiated cells are integral to the pathology of IPF, with macrophages, in particular, being identified as pivotal contributors to disease progression. This review aims to elucidate the primary pathways involved in monocyte recruitment to the lungs during IPF and to investigate the crucial roles that monocytes play in the disease's pathogenesis. This review aims to establish a foundation for novel therapeutic strategies targeting monocytes, thereby facilitating early detection and intervention in IPF.

Keywords Idiopathic pulmonary fibrosis · Monocytes · Macrophages · Chemokines · Extracellular matrix

Abbreviations

AEC-I	Type I alveolar epithelial cells
AEC-II	Type II alveolar epithelial cells
AMs	Alveolar macrophages
ApoE	Apolipoprotein E
BALF	Bronchoalveolar lavage fluid
BLM	Bleomycin
CCL2	Chemokine ligand 2
CSF1	Colony-stimulating factor 1
CTGF	Connective tissue growth factor
DAMPs	Damage-associated molecular patterns
ECM	Extracellular matrix
EMT	Epithelial-mesenchymal transition
EndMT	Endothelial-mesenchymal transition
FGF	Fibroblast growth factor
FMT	Fibroblast-to-myofibroblast transition
IFN-γ	Interferon-γ
IGF	Insulin-like growth factor
IL-1β	Interleukin-1β
IL-4	Interleukin-4
ILC2s	Type 2 innate lymphoid cells
IMs	Interstitial macrophages
LRP1	Lipoprotein receptor-related protein 1
MCP-1	Monocyte chemoattractant protein-1
MCPIP1	MCP-1 induced protein 1
M-CSF	Macrophage colony-stimulating factor
MHC	Major histocompatibility complex
MMPs	Matrix metalloproteinases
MMs	Monocyte-derived macrophages
Mo-AMs	Monocyte-derived alveolar macrophages

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Mo-IMs	Monocyte-derived interstitial macrophages
MSCs	Mesenchymal stem cells
NAMPT	Nicotinamide phosphoribosyltransferase
OPN	Osteopontin
PAMPs	Pathogen-associated molecular patterns
PDGF	Platelet-derived growth factor
IPF	Idiopathic pulmonary fibrosis
TGF-β1	Transforming growth factor-beta 1
TLRs	Toll-like receptors
TRMs	Tissue-resident macrophages

Introduction

Idiopathic pulmonary fibrosis (IPF) is a progressive and incurable interstitial lung disease characterized by the excessive accumulation of extracellular matrix (ECM), resulting in the disruption of lung tissue architecture and a decline in pulmonary function. Alveolar collapse can exacerbate damage to pulmonary venous vessels, potentially leading to complications such as pulmonary hypertension [1, 2]. Currently, the prevalence of IPF is estimated at approximately 50 per 100,000 individuals, with a higher incidence observed in males with a history of smoking. However, due to the escalating severity of environmental pollution and other contributing factors, the number of individuals affected by IPF is projected to surpass 1.5 billion by 2050, with an ongoing annual increase [3]. The median survival time for individuals with IPF is approximately three years. Aside from lung transplantation for patients in advanced stages of the disease, there are currently no effective treatment options available. Despite the approval of the anti-fibrotic drugs Pirfenidone and Nintedanib by the U.S. Food and Drug Administration, which aid in delaying the progression of IPF, these medications do not restore lung tissue function and are associated with adverse effects that limit their clinical utility [4, 5]. Consequently, there is an urgent need for the development of therapeutic agents capable of reversing or halting the progression of IPF, with the aim of enhancing patient survival rates and quality of life.

The pathogenesis of IPF involves multiple cell types, including epithelial cells, macrophages, fibroblasts, and monocytes. It is widely accepted that the condition is primarily initiated by injury to alveolar epithelial cells. These damaged epithelial cells release substantial amounts of pro-inflammatory factors and chemokines, which recruit immune cells such as monocytes and macrophages to the injury site. These immune cells, in turn, produce significant quantities of chemokines and inflammatory mediators, thereby perpetuating a vicious cycle [6, 7]. Monocyte-derived macrophages (MMs) are known to secrete pro-fibrotic factors that facilitate the differentiation of fibroblasts into myofibroblasts. These myofibroblasts subsequently produce excessive ECM, leading to the disruption of lung tissue architecture and function, thereby advancing the progression of IPF [8, 9]. Monocytes, as precursor cells to macrophages, exhibit significant capabilities in immune surveillance, antigen presentation, and phagocytosis. During the course of IPF, monocytes can differentiate into interstitial macrophages (IMs) and alveolar macrophages (AMs). The peripheral blood monocyte count has been identified as a robust biomarker indicative of poor prognosis in IPF. Current research suggests that circulating MMs, rather than tissue-resident macrophages (TRMs), are critical drivers of IPF pathogenesis [10, 11]. This review aims to synthesize key pathways involving monocytes and macrophages in the pathogenesis of IPF, offering novel insights and perspectives for the prediction and treatment of this disease.

Idiopathic pulmonary fibrosis

The pathological hallmark of IPF is characterized by the excessive production and aberrant deposition of ECM, resulting in the disruption of lung tissue architecture and subsequent loss of function. A variety of therapeutic strategies have been explored for IPF, including interventions targeting pro-fibrotic signaling pathways, cellular senescence, the immune system, metabolic processes, gene therapy, and stem cell therapy. Therefore, the prominent inflammatory and fibrotic components of IPF progression make anti-inflammatory, immunomodulatory, and anti-fibrotic agents logical therapeutic targets. In the context of anti-fibrotic therapy, transforming growth factor-beta 1 (TGF-β1) plays a pivotal role as an inducer of several fibrogenic factors, including connective tissue growth factor (CTGF), fibroblast growth factor (FGF), insulin-like growth factor (IGF), and platelet-derived growth factor (PDGF). Although inhibitors targeting TGF-β1, such as BG00011 and GSK3008348, were developed, they have been withdrawn from clinical development due to severe adverse reactions [12, 13]. In the context of immunotherapy, current research focuses on the recruitment and differentiation of macrophages and T cells. The polarization of macrophages towards the M2 phenotype and the differentiation of T cells into TH2, TH17, and $CD8^+$ subsets facilitate the progression of IPF. Conversely, an increase in tissue-resident $CD4^+$ T cells and TH1 cells exerts a protective effect [14, 15]. Recent studies have demonstrated that an elevated number of circulating monocytes within lung tissue serves as a predictor for the progression of IPF [16]. The pathogenesis of IPF involves multiple distinct cell types within lung tissue, including epithelial cells, endothelial cells, macrophages, monocytes, fibroblasts, and

neutrophils. These cells engage in complex interactions that collectively drive the advancement of IPF (Fig. 1).

Alveolar epithelial cells in IPF

The alveolar epithelium within lung tissue is categorized into two distinct subtypes: type I and type II alveolar epithelial cells (AEC-I and AEC-II). AEC-I cells are primarily responsible for facilitating gas exchange, while AEC-II cells function as progenitor cells for AEC-I, differentiating into them to sustain alveolar function. Furthermore, AEC-II cells play a crucial role in maintaining the homeostasis of the alveolar microenvironment through the secretion of surfactant. In the context of IPF, AEC-II cells are central players, critically implicated in initiating the fibrotic process. Following repeated external injury, AEC-II cells exhibit alterations such as telomere shortening, mitochondrial dysfunction, and cellular senescence, which compromise the gas exchange capacity of the alveolar epithelium, resulting in clinical manifestations such as dyspnea, cough, and fatigue in patients. Notably, cellular senescence of AEC-II cells is an early hallmark of IPF. Subsequently, these cells secrete significant amounts of pro-inflammatory cytokines and chemoattractants, including interleukin-1β (IL-1β), IL-6, TNF-α, and CCL-2 [17]. Pro-inflammatory mediators facilitate the recruitment of immune cells, including macrophages, monocytes, neutrophils, and lymphocytes, from both pulmonary tissue and the circulatory system to the site of lung injury. The subsequent recruitment and activation of these immune cells at the injury locus further exacerbate the inflammatory cascade through the production of substantial amounts of pro-inflammatory cytokines and chemoattractants [18, 19]. Additionally, both the recruited immune cells and the chronically injured alveolar epithelial cells (AECs) can secrete pro-fibrotic factors, such as IL-1β and TGF-β1, which stimulate the extensive proliferation of fibroblasts and their differentiation into myofibroblasts. The ECM deposited by these cells results in the formation of fibrous connective tissue. Although this deposition aids in tissue repair, the regenerated lung tissue lacks the original architecture and function, thereby facilitating the progression of IPF, in accordance with the "epithelial injury hypothesis" [20]. Research has shown that in bleomycin (BLM)-induced IPF, chemoattractants such as CCL2 and CCL12, released by injured type II alveolar epithelial cells (AEC-IIs), can recruit monocytes, which subsequently differentiate into macrophages and contribute to IPF [21, 22]. Furthermore, during IPF, AEC-IIs can undergo epithelial-mesenchymal transition (EMT), acquiring mesenchymal cell characteristics, which enhances ECM expression and thereby promotes the fibrotic process.

Fibroblasts in IPF

Fibroblasts, the primary non-immune cells involved in fibrosis, play a pivotal role in IPF. In the context of IPF, fibroblasts differentiate into myofibroblasts and secrete substantial amounts of ECM, thus facilitating the progression

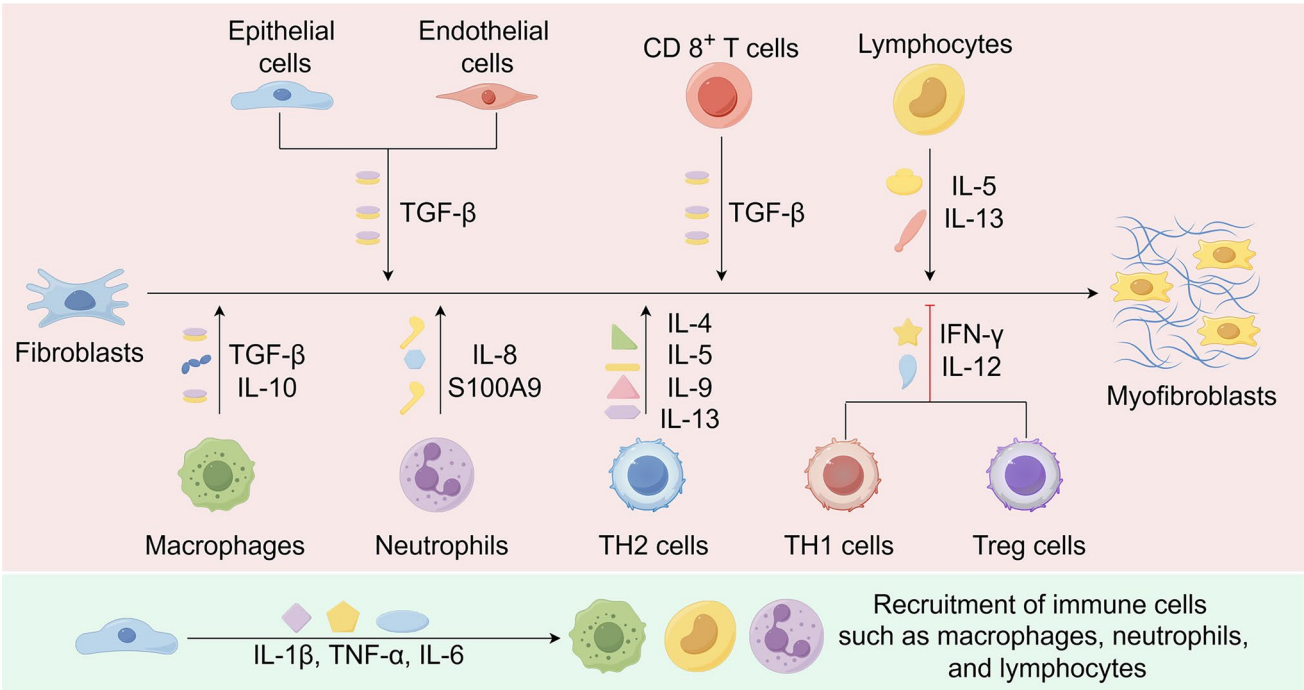

Fig. 1 Roles of different cells in idiopathic pulmonary fibrosis. This diagram was drawn by Figdraw (www.figdraw.com)

of IPF. The fibroblast focus refers to clusters of highly proliferative and activated fibroblasts and myofibroblasts within the lung interstitium. These cells exhibit increased proliferative capacity and resistance to apoptosis, representing a hallmark feature of IPF and an indicator of poor prognosis. Fibroblasts and myofibroblasts are the principal cells responsible for ECM secretion. Research has demonstrated that sustained elevation in the expression of S100a4, a fibroblast-specific protein, is predictive of the transformation of fibroblasts into myofibroblasts and the progression of fibrosis [23, 24]. Under typical injury conditions, fibroblasts undergo apoptosis and facilitate epithelial reconstitution upon completing their functional role, thereby preserving the integrity of the affected site. In contrast, in IPF, the persistent activation of myofibroblasts and subsequent cellular senescence result in the continuous accumulation of these cells and confer resistance to apoptosis. This leads to impaired epithelial reconstitution and dysfunctional tissue healing, thereby exacerbating the progression of IPF. It is important to note that the origins of fibroblast and myofibroblast populations in IPF are heterogeneous. Beyond resident fibroblasts and myofibroblasts, processes such as endothelial-mesenchymal transition (EndMT), EMT, pericyte transition, and macrophage transition contribute to the formation of these cellular clusters. TGF-β1, a pivotal cytokine in IPF, plays a significant role in inducing apoptosis of alveolar type 2 cells, EMT, fibroblast-to-myofibroblast transition (FMT), and the deposition of interstitial ECM and collagen [25, 26]. In addition to the ECM, the excessive accumulation of collagen is a notable feature of IPF. Collagen comprises approximately 20% of lung weight and has the capacity to cross-link with fibroblasts, thereby affecting the mechanical strength of the ECM. During the progression of IPF, both FMT and epithelial-to-mesenchymal transition (EMT) can enhance collagen secretion. Macrophages and AEC-II serve as the principal sources of TGF-β1, although endothelial cells and $CD8^+$ T cells also contribute to its release. TGF-β1 facilitates IPF by recruiting fibroblasts to sites of injury, inducing their differentiation into myofibroblasts, and promoting ECM production. Moreover, in the context of IPF, tumor necrosis factor-alpha (TNF-α), PDGF, and various chemoattractants further contribute to the progression of fibrosis.

Macrophages in IPF

Macrophages are integral to the recognition and clearance of foreign pathogens and damage-associated molecular patterns (DAMPs). These cells exhibit high plasticity, allowing them to transition between distinct polarization phenotypes, specifically M1 and M2 macrophages, in response to various cytokine stimuli. M1 macrophages are characterized by their pro-inflammatory nature, releasing inflammatory mediators such as TNF-α and IL-1β, and demonstrating significant phagocytic capabilities [27, 28]. The polarization of M1 macrophages is predominantly driven by the JAK-STAT or TLR-NF-κB signaling pathways, which are associated with the upregulation of genes such as CD86, TNF-α, and MHC-II [29, 30]. While M1 macrophages are primarily recognized for their pro-inflammatory roles, TNF-α and IL-1β also function as pro-fibrotic agents, contributing to IPF by inducing EMT and fibroblast-to-myofibroblast transformation, ultimately resulting in parenchymal lung injury. Research indicates that sustained inflammatory responses can facilitate fibrotic progression. During the initial stages of IPF, there is a notable shift of AMs towards the M1 phenotype, which contributes to the fibrotic process through the release of TGF-β1 and the activation of Smad2/3 pathways [31, 32]. Conversely, M2 macrophages are known for their anti-inflammatory properties and significant involvement in tissue repair. It is well-established that M2 macrophages secrete cytokines such as IL-4, IL-13, IL-10, and TGF-β1, which facilitate the proliferation and activation of fibroblasts, myofibroblasts, and additional M2 macrophages. An excessive differentiation of M2 macrophages is considered a key factor in the progression of fibrosis. The polarization of M2 macrophages can induce the release of TGF-β1 and Smad2/3, as well as promote EMT [33]. Macrophage polarization is recognized as a dynamic process, where the interaction between excessively released cytokines and macrophages results in an imbalance of M1/M2 polarization phenotypes. Consequently, targeting macrophage polarization may offer a promising therapeutic approach for the treatment of IPF in the future.

Neutrophils in IPF

Neutrophils, prevalent immune cells within the bloodstream, are recruited to sites of tissue injury where they initiate inflammatory responses, thereby playing an essential role in tissue repair processes. In individuals diagnosed with IPF, approximately 70% present with a neutrophil percentage exceeding 3% in bronchoalveolar lavage fluid (BALF), while more than 50% exhibit levels surpassing 5% in BALF [34, 35]. It is important to note that these measurements are expressed as percentages rather than absolute counts. Research has demonstrated a correlation between elevated neutrophil percentages in BALF and both the severity of IPF and increased mortality rates [34, 36]. While the functional activity of neutrophils may be more critical than their sheer numbers, the precise role of neutrophils in the pathogenesis of IPF remains inadequately understood and warrants further investigation.

Other cells in IPF

In the context of IPF, T cells, B cells, and natural killer (NK) cells are implicated in the pathogenesis, as outlined in references [37–39]. It is crucial to recognize that the primary cellular constituents involved in IPF include alveolar epithelial cells, fibroblasts, and macrophages. Alveolar epithelial cells are the initial targets of lung tissue injury, fibroblasts are pivotal in the secretion of ECM, and macrophages are central to receiving injury signals and inducing ECM production. While the mechanisms involving alveolar epithelial cells and fibroblasts are relatively well-characterized, the role of macrophages remains a subject of debate. Lung tissue resident macrophages are primarily composed of AMs and IMs, both of which play roles in immune surveillance and defense. During the progression of IPF, TRMs are supplanted by circulating monocytes. These monocytes differentiate into macrophages upon tissue infiltration and secrete substantial amounts of pro-inflammatory mediators, thereby exacerbating lung tissue damage. Notably, IMs are less prevalent compared to AMs. In pathological conditions, Mo-IMs may contribute to tissue repair and the advancement of IPF. Studies have demonstrated that IMs predominantly exhibit an M2 phenotype, and targeted inhibition of these cells can impede the progression of IPF [40, 41]. MMs are crucial in the pathogenesis of IPF, with an increase in their relative abundance serving as a promising biomarker for the disease. During the course of IPF, TRMs are a major source of chemoattractants, facilitating the recruitment of circulating monocytes from the bloodstream by releasing substantial quantities of these factors. Additionally, cytokines such as TGF-β1 and interleukin-4 (IL-4), secreted by macrophages, can recruit circulating monocytes and promote their differentiation into M2 macrophages. These M2 macrophages subsequently release more TGF-β1 and IL-4, perpetuating a detrimental cycle that exacerbates the progression of IPF.

Monocyte function

Monocytes were initially identified in 1926 and were regarded as immature constituents of the reticuloendothelial system [42]. Originating primarily from myeloid precursors, specifically hematopoietic stem and progenitor cells within the bone marrow, monocytes are extensively distributed throughout the bloodstream, accounting for 2–8% of the total white blood cell population. These cells are integral to the maintenance of immune homeostasis and the defense against pathogenic invasion. Their functions encompass immune surveillance, recruitment and migration to sites of tissue injury, phagocytosis of pathogens, antigen processing and presentation, as well as the modulation of both pro-inflammatory and anti-inflammatory responses [43, 44]. Monocytes exhibit significant capabilities in immune surveillance and antigen processing/presentation. As pivotal components of immune surveillance, they possess the ability to directly recognize exogenous antigens, such as pathogens, and are responsive to inflammatory mediators and chemoattractants released during tissue damage. In the context of adaptive immune responses, monocytes are responsible for recognizing exogenous antigens, internalizing them through phagocytosis, and processing them into small peptides. These peptides are subsequently presented to T cells via major histocompatibility complex (MHC) molecules, thus initiating T-cell-mediated immune responses [45]. Additionally, monocytes contribute to T cell activation by releasing cytokines and co-stimulatory molecules. It is important to note that dendritic cells and macrophages, which differentiate from monocytes, maintain their capacities for immune surveillance and antigen presentation. These cells exhibit specialized antigen-presenting functions across various tissues and collaborate to enhance adaptive immune responses within these tissues. For example, in pulmonary tissue, IMs are capable of presenting soluble antigens via MHC-II molecules, leading to the induction of regulatory T cells. In adipose tissue, macrophages facilitate the proliferation of $CD4^+$ T cells and the production of interferon-γ (IFN-γ) through MHC-II molecules, thereby playing crucial roles in local immune responses [46].

Monocytes are capable of receiving signals, including pro-inflammatory factors and chemoattractants, that are released during tissue injury. These signals facilitate the transmigration of monocytes across the vascular wall, leading to their recruitment at the site of injury. This process is crucial for maintaining immune homeostasis and promoting tissue repair. Research indicates that the recruitment of circulating monocytes to sites of infection or injury is mediated by Toll-like receptors (TLRs) [47, 48]. Activation of TLRs stimulates the release of the chemokine CCL-2 from bone marrow cells, which subsequently binds to the CCR2 receptor on monocytes, prompting their migration to injured tissues. In the context of tumor-related diseases, CCL2 enhances the recruitment and activation of monocytes at the lesion site through its interaction with CCR2, thereby facilitating tumor cell dissemination and growth. Similar mechanisms have been observed in studies of cardiac, pulmonary, and other tissue-related pathologies. Notably, once recruited to damaged tissues, monocytes have the potential to differentiate into macrophages and dendritic cells, further contributing to the regulation of immune responses. Both circulating monocytes and macrophages demonstrate significant phagocytic capabilities upon infiltrating injured tissues, playing a crucial role in the clearance of foreign

pathogens, cellular debris, and particulate matter. During the phagocytosis of pathogens, monocytes release substantial quantities of inflammatory mediators to modulate immune responses at the site of injury. Additionally, monocytes can respond to inflammatory cues by releasing migrasomes enriched with inflammatory factors, which rapidly accumulate at sites of inflammation to further regulate immune responses. Under pathological conditions, monocytes not only secrete pro-inflammatory factors that promote M1 macrophage polarization but also release anti-inflammatory factors that facilitate M2 macrophage polarization, thereby modulating both pro-inflammatory responses and tissue repair processes. Importantly, monocytes are recruited from the circulation to injured tissues and can recirculate back into the bloodstream by traversing the vascular endothelium or migrating into lymph nodes. Moreover, monocytes may return to the bone marrow to support the fundamental physiological functions of the organism. Activation of TLRs stimulates the release of the chemokine CCL-2 from bone marrow cells, which subsequently binds to the CCR2 receptor on monocytes, prompting their migration to injured tissues [49]. In the context of tumor-related diseases, CCL2 enhances the recruitment and activation of monocytes at the lesion site through its interaction with CCR2, thereby facilitating tumor cell dissemination and growth [50, 51]. Similar mechanisms have been observed in studies of cardiac, pulmonary, and other tissue-related pathologies [52, 53]. Notably, once recruited to damaged tissues, monocytes have the potential to differentiate into macrophages and dendritic cells, further contributing to the regulation of immune responses [54]. Both circulating monocytes and macrophages demonstrate significant phagocytic capabilities upon infiltrating injured tissues, playing a crucial role in the clearance of foreign pathogens, cellular debris, and particulate matter [55]. During the phagocytosis of pathogens, monocytes release substantial quantities of inflammatory mediators to modulate immune responses at the site of injury [56, 57]. Additionally, monocytes can respond to inflammatory cues by releasing migrasomes enriched with inflammatory factors, which rapidly accumulate at sites of inflammation to further regulate immune responses. Under pathological conditions, monocytes not only secrete pro-inflammatory factors that promote M1 macrophage polarization but also release anti-inflammatory factors that facilitate M2 macrophage polarization, thereby modulating both pro-inflammatory responses and tissue repair processes [58, 59]. Importantly, monocytes are recruited from the circulation to injured tissues and can recirculate back into the bloodstream by traversing the vascular endothelium or migrating into lymph nodes. Moreover, monocytes may return to the bone marrow to support the fundamental physiological functions of the organism [60].

Monocytes can be categorized into three subsets based on surface receptor expression: $CD14^{++}CD16^{-}$ (classical), $CD14^{+}CD16^{++}$ (non-classical), and $CD14^{++}CD16^{+}$ (intermediate) [61, 62]. Each subset fulfills distinct roles under both homeostatic and pathological conditions. Classical monocytes, characterized by high expression of CCR2, are primarily involved in tissue infiltration, phagocytosis, and pro-inflammatory functions. They facilitate migration and adhesion to injury sites and are pivotal in the release of chemoattractants and pro-inflammatory cytokines. Intermediate monocytes, which express high levels of MHC-II, are predominantly associated with antigen processing and presentation, specifically presenting a broad range of antigens to $CD4^{+}$ T cells, and are involved in monocyte activation. Non-classical monocytes, expressing high levels of MHC-I, specialize in vascular patrolling and immune surveillance. They present viral peptides to $CD8^{+}$ T cells and are important in wound healing. In response to inflammatory stimuli, classical monocytes primarily secrete substantial quantities of cytokines, including IL-6, IL-8, CCL-2, and CCL-3. Intermediate monocytes are the principal producers of pro-inflammatory cytokines such as TNF-α and IL-1β, while non-classical monocytes are capable of secreting TNF-α, IL-1β, and CCL3, among other cytokines. Notably, all three monocyte subsets possess the ability to secrete the pro-inflammatory cytokines IL-1β, IL-6, and TNF-α. Additionally, intermediate and non-classical monocytes can be stimulated to produce reactive oxygen species (ROS), whereas classical monocytes demonstrate a limited capacity for ROS generation, a phenomenon that may be associated with variations in CD14 expression.

Monocyte differentiation

Monocytes are responsive to chemoattractant signals and are subsequently recruited to sites of inflammation or injury. Upon arrival at specific tissue environments, monocytes undergo differentiation into macrophages or dendritic cells, guided by local environmental cues [63, 64]. Macrophages, as pivotal components of the innate immune system, play a fundamental role in immune defense. The differentiation of monocytes into macrophages is mediated in an NF-κB-dependent manner under the influence of macrophage colony-stimulating factor (M-CSF). Dendritic cells, on the other hand, recognize pathogen-associated molecular patterns (PAMPs) or DAMPs and present these antigens to B lymphocytes and T lymphocytes, thereby activating their effector functions and serving as essential intermediaries between innate and adaptive immunity. Alterations in monocyte morphology and immunophenotype, which regulate innate immune responses, are implicated in various

pulmonary inflammatory conditions, lung failure, and even mortality.

Monocytes, upon responding to inflammatory signals, migrate to local tissues where they differentiate into macrophages, thus constituting a significant source of tissue macrophages. These macrophages exhibit extensive phagocytic, bactericidal, and secretory functions. Through the process of phagocytosis, macrophages effectively eliminate foreign antigens, pathogenic microorganisms, and cellular debris, subsequently utilizing these materials as metabolic substrates and energy sources to sustain organismal homeostasis. When foreign pathogen signals are recognized via pattern recognition receptors or cytokines released by other host cells, macrophages produce substantial quantities of reactive oxygen species and lysosomal enzymes, which are detrimental to invaders such as bacteria. Furthermore, macrophages secrete significant amounts of inflammatory cytokines (e.g., TNF-α, IL-1β) and chemokines (e.g., CCL2) through the NF-κB signaling pathway, thereby modulating immune and inflammatory responses within tissues [65, 66]. These cytokines also play crucial roles in enhancing vascular permeability and recruiting additional immune cells. Moreover, macrophage functions exhibit variability across different tissues, as tissue-specific signals direct macrophages to adapt to distinct microenvironments. In conclusion, macrophages are essential components of the immune system, performing a variety of functions, including phagocytic, bactericidal, and secretory activities, which are vital for maintaining homeostasis and defending against pathogens [67, 68]. Macrophages demonstrate significant plasticity, with the ability to interconvert between different subtypes. Functionally, macrophages are broadly classified into M1 macrophages (classically activated, primarily exerting pro-inflammatory effects) and M2 macrophages (alternatively activated, primarily mediating anti-inflammatory effects) [69–71]. Some studies suggest that M1 macrophages may inhibit fibrosis and wound healing, whereas M2 macrophages may promote these processes [72–75]. With advancements in macrophage research, numerous macrophage subpopulation classification methods have been proposed, rendering the simplistic M1/M2 functional dichotomy inadequate for distinguishing disease-specific macrophage subsets. Notably, macrophages are the most abundant immune cell type in lung tissue. Lung macrophages can be categorized based on their origin into MMs, AMs, and IMs [76, 77]. MMs serve to supplement the other two populations. AMs are primarily responsible for recognizing and eliminating exogenous pathogens, regulating pro-inflammatory responses, and facilitating tissue repair [78–80]. IMs play a crucial role in modulating pro-inflammatory responses and antigen presentation [81]. Presently, the role of monocytes in IPF remains underexplored. Nonetheless, emerging research indicates that monocyte count may be a significant prognostic marker for mortality and adverse outcomes in IPF. In murine models of IPF, newly recruited circulating MMs have been shown to contribute to the disease's progression [82, 83].

Under pathological conditions, monocyte-derived dendritic cells constitute a major source of dendritic cells. Inflammatory factors facilitate not only the recruitment and migration of monocytes to damaged tissues but also their differentiation into dendritic cells, which exhibit robust capabilities for antigen uptake, processing, and presentation, thereby serving as a crucial link between innate and adaptive immunity. After antigen uptake and processing, dendritic cells activate adaptive immune responses through interactions with T cells. Research indicates that monocytes in vivo can initiate differentiation into dendritic cells through interactions with specific cytokines, such as GM-CSF and IL-4, which activate multiple signaling pathways, including MAPK, JAK/STAT, NF-κB, and PI3K [84–87]. Moreover, in inflammatory environments, monocytes can differentiate into regulatory dendritic cells through direct contact with regulatory T cells and helper T cells [88]. This process is vital for maintaining organismal homeostasis and presents potential for immunotherapeutic applications. Within the tumor microenvironment, monocyte-derived dendritic cells play a pivotal role in activating adaptive immune responses in T cells through cross-presentation mechanisms and modulating immune-related functions via lipid metabolism, thereby constituting a crucial strategy in antitumor immune regulation [89, 90]. The migration and positioning of these cells are critical determinants of their functional efficacy at sites of inflammation and infection. A comprehensive investigation into the biological characteristics and functional mechanisms of dendritic cells may yield novel insights and approaches for immunotherapy, as well as for the prevention and control of diseases. Further elucidation of the mechanisms governing monocyte differentiation into dendritic cells will enhance our understanding of immune system dynamics and offer new perspectives for the treatment of related diseases.

Idiopathic pulmonary fibrosis and monocytes

In the pathogenesis of IPF, the pathological process is primarily delineated into an acute inflammatory phase and a chronic fibrotic phase. The acute phase is marked by a rapid influx of neutrophils, a decrease in TRMs, and the release of abundant inflammatory mediators. In contrast, the chronic phase is characterized by a transition from innate immunity to adaptive immunity, with MMs and T cells predominating, alongside the secretion of numerous pro-fibrotic

mediators. Current therapeutic interventions for IPF predominantly focus on decelerating disease progression rather than reversing fibrosis, highlighting the critical need for the identification of early disease biomarkers. Although several potential biomarkers have been proposed, none have received endorsement from the European Respiratory Society or the American Thoracic Society for clinical application. Dysregulated immune responses are pivotal in driving ECM accumulation across various IPF-related disorders.

The critical role of monocytes in idiopathic pulmonary fibrosis

As early as 1991, research identified a significant increase in monocyte content, the precursor cells of macrophages, within lung tissue during instances of IPF [91]. Employing immunocytochemical staining with the Ki-M antibody series to delineate the differentiation stages of monocyte/macrophage subsets, it was observed that a greater proportion of AMs positive for Ki-M6 and Ki-M8 displayed an immunophenotype resembling that of monocytes. In the context of IPF, monocytes present in lung tissue are capable of differentiating into AMs, thereby promoting the release of free radicals and facilitating the progression of the disease. In a silica-induced mouse model of IPF, there was a marked upregulation in the proportion of monocytes within lung tissue, particularly those expressing CXCL10 and MMP14 [92]. The accumulation of MMs constitutes a crucial supplement to tissue macrophages and plays a pivotal role in advancing IPF. Under homeostatic conditions, TRMs in the lung are able to clear foreign pathogens and cellular debris without eliciting an inflammatory response. In contrast, MMs, which are recruited and differentiate during the progression of IPF, exhibit both pro-inflammatory and pro-fibrotic effects in the process of pathogen clearance (Fig. 2). Importantly, upon migration from the bone marrow to the injured lung tissue, monocytes have the capacity to differentiate not only into pro-fibrotic macrophages but also into fibroblasts. Elevated levels of MMs and circulating fibrocytes within lung tissue, alongside increased concentrations of CCL18 (produced by AMs) in the bloodstream, may serve as predictive biomarkers for IPF [93]. Research indicates that during lung tissue disease, monocytes recruited to sites of injury can facilitate tissue repair; however, the persistence of pathological processes can lead to ongoing tissue remodeling, ultimately resulting in IPF. In cases of IPF, there is a positive correlation between peripheral blood $CD14^+$ monocytes and both poor prognosis and increased mortality risk. A monocyte count of $\geq 0.95 \times 10^9$/L is associated with a high risk of mortality, and when using a threshold of 0.67×10^9/L, patients with monocyte counts above this level are at an elevated risk of mortality [94]. Monocytes serve as potential biomarkers for predicting poor prognosis in IPF. Upon recruitment to sites of lung injury, these cells can differentiate into macrophages. The depletion of Mo-AMs has been shown to effectively decelerate the progression of IPF. In patients with COVID-19, there is an observed increase in the proportion of MMs, accompanied by elevated expression of chemokines such as CCL7, CCL8, and CCL18, which are closely associated with the development of IPF [95, 96]. Chemokines, a subclass of cytokines, enhance local pro-inflammatory and pro-fibrotic responses by recruiting leukocytes and lymphocytes. Research indicates that MCP-1/CCR2, CXCL12/CXCR4, CXCL4, and CCL1 are implicated in the pathogenesis of fibrosis-related diseases [97–99]. The interaction between GM-CSF/GM-CSFR and MCP1/CCR2 is pivotal in macrophage proliferation, migration, and differentiation [100]. Ex vivo studies demonstrate that GM-CSF and MCP-1 can induce macrophage differentiation towards the M2 phenotype, whereas neutralizing antibodies against GM-CSF can significantly reduce M2 macrophage levels in the liver tissue of mice with hepatitis. In a murine model of IPF induced by SiO_2, MCP-1 Induced Protein 1 (MCPIP1), a critical downstream effector of MCP-1, predominantly facilitates macrophage autophagy, thereby advancing macrophage apoptosis during IPF [101]. The tumor suppressor protein p53 sustains the regulatory function of MCPIP1 in autophagy. MCPIP1 activates p53 to orchestrate macrophage autophagy, apoptosis, and activation, which in turn modulates the proliferation and migration of lung fibroblasts, consequently contributing to the progression of IPF. Silica exposure prompts circulating monocytes to infiltrate lung tissue in a CCR2-dependent manner, where they differentiate into macrophages; the proportion and number of these MMs progressively increase throughout the disease course. The surface of MMs is enriched with innate immune sensing pathways, including Toll-like receptors (TLR3, TLR7/8, TLR9), which are capable of recognizing exogenous antigens such as SARS-CoV-2 and BLM, thereby promoting differentiation towards a pro-fibrotic phenotype [102–104]. The expression of mast cell-expressed membrane protein 1 (MCEMP1) is significantly elevated in classical monocytes and alveolar macrophages in cases of idiopathic pulmonary fibrosis (IPF) [105]. MCEMP1 plays a crucial role in modulating TGFβ-mediated migration and the transformation of monocytes into Mo-AMs. Its expression level is strongly positively correlated with increased mortality risk in IPF. In the context of pulmonary fibrosis, a signaling circuit is established through the interaction between CSF1R, expressed by Mo-AMs, and M-CSF, secreted by fibroblasts. This interaction promotes the recruitment of Mo-AMs and facilitates the differentiation and proliferation of fibroblasts. The proper functioning of the epithelial repair mechanism

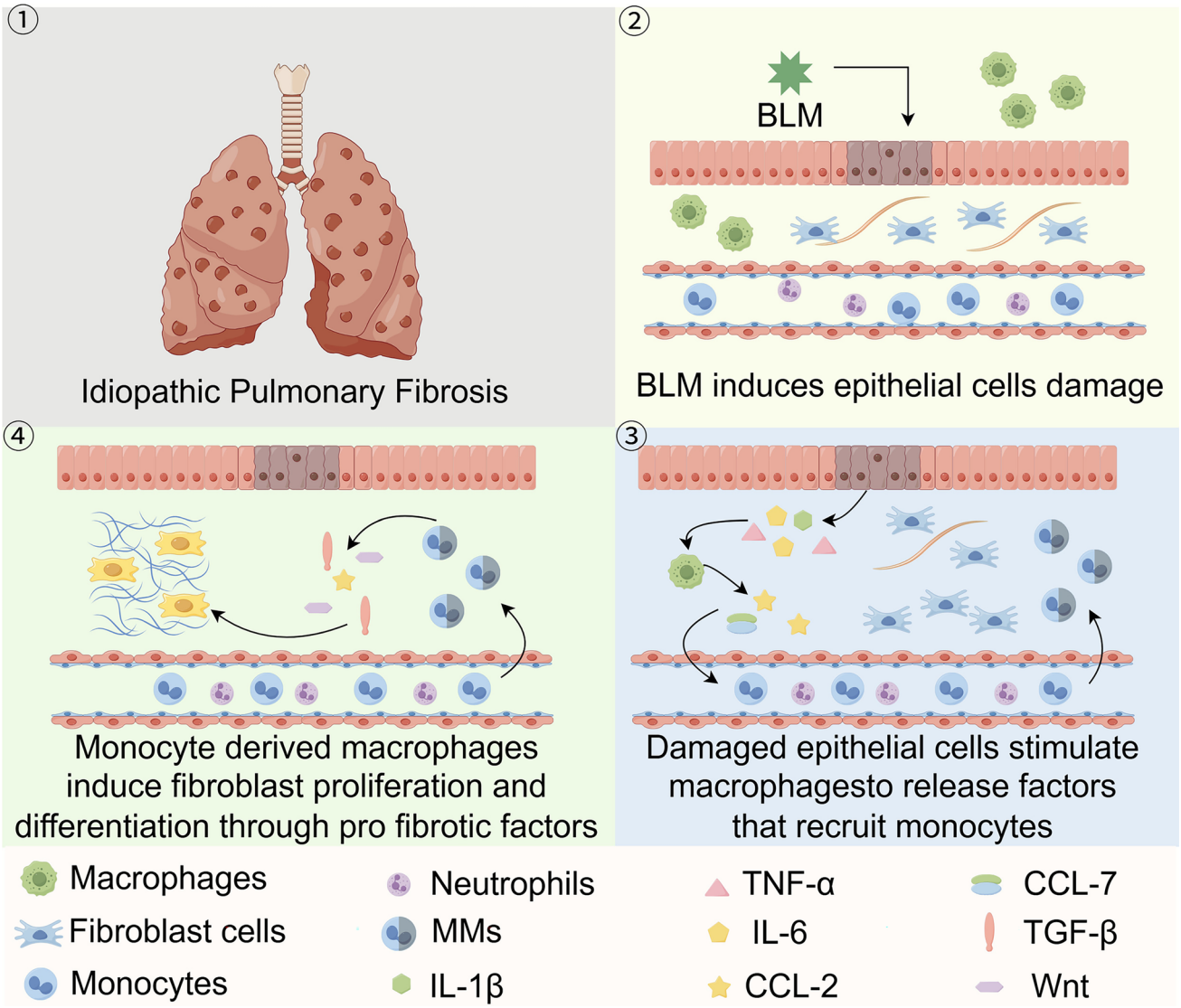

Fig. 2 Key roles of monocytes in idiopathic pulmonary fibrosis. This diagram was drawn by Figdraw (www.figdraw.com)

encourages the differentiation of Mo-AMs into tissue-resident alveolar macrophages or triggers their apoptosis via GM-CSF signaling. This process effectively disrupts the interaction between alveolar macrophages and fibroblasts. ISRIB, a small molecule that reinstates protein translation following the activation of the integrated stress response (ISR), enhances the expression of eukaryotic initiation factor 2B (EIF2B) proteins during ISR activation. It facilitates the differentiation of alveolar type II epithelial cells into alveolar type I epithelial cells, thereby aiding in the repair of epithelial injury and reducing epithelial apoptosis [106]. This mechanism mitigates the recruitment of pro-fibrotic Mo-AMs and alleviates pulmonary fibrosis. Moreover, single-cell and cytokine analyses of the alveolar space in patients experiencing post-acute sequelae of COVID-19 have demonstrated a positive correlation between the abundance of Mo-AMs, the levels of the monocyte chemoattractant CCL2, and the severity of pulmonary fibrosis [95]. Mo-AMs have been identified as potential biomarkers indicative of impaired lung tissue repair, with their targeted depletion shown to significantly enhance pathological outcomes in pulmonary fibrosis. The expedited regeneration of alveolar epithelial cells disrupts the recruitment process of Mo-AMs, facilitates their differentiation into tissue-resident alveolar macrophages, and attenuates the advancement of pulmonary fibrosis.

Regulation of monocyte recruitment in idiopathic pulmonary fibrosis—the chemokine pathway

Monocytes and MMs are recognized as the primary amplifying cells during the initial stages of IPF progression,

playing a pivotal role in the early phase of the disease. In the context of IPF, endothelial cells within lung tissue facilitate endothelial-to-mesenchymal transition and augment endothelial permeability through the MMP19-regulated SDF1/CXCR4 pathway, thereby promoting monocyte infiltration into the lung tissue [107]. The recruitment of monocytes and macrophages to sites of lung injury is primarily governed by the regulation of chemoattractants, such as CCL2. The interaction between the monocyte chemoattractant protein 1 receptor (CCR2) and the C–C motif chemokine ligand 2 (CCL2) is a critical factor in regulating macrophage trafficking. CCR2 is predominantly expressed on the surface of monocytes and serves as a significant surface marker for these cells. CCL2 is secreted by a variety of cell types, including monocytes, macrophages, fibroblasts, epithelial cells, and endothelial cells. $CD11b^{hi}$ macrophage-like monocytes (MMs) are significant contributors to the production of pro-fibrotic chemokines, notably through the upregulation of CCL2 and CCL24, during the progression of IPF [108]. These chemokines facilitate the recruitment, proliferation, and secretion of ECM components by fibrocytes, fibroblasts, and myofibroblasts, thereby exacerbating fibrotic development. The interaction between CCL2 and its receptor CCR2 is crucial for the recruitment of macrophages and circulating monocytes to sites of lung injury. Once recruited, these monocytes can differentiate into pro-fibrotic interstitial and AMs, which secrete substantial amounts of TGF-β1, further advancing IPF. Empirical studies have shown that in BLM-induced mouse models of IPF, mice with genetic deficiencies in either CCL2 or CCR2 exhibit a marked reduction in macrophage infiltration, along with decreased expression of MMP-2 and MMP-9. This reduction correlates with lower lung hydroxyproline content and improved lung function in fibrotic tissue. Notably, elevated levels of $CCR2^{+}$ monocytes persist in the lung tissues of patients with end-stage IPF. Both monocytes and MMs play a critical role in the advancement of IPF. The targeted reduction of $CCR2^{+}$ cells has demonstrated potential in alleviating liver and renal fibrosis. Similarly, the depletion of $CD11b^{+}$ MMs has been shown to effectively mitigate IPF in BLM-induced mouse models. In cases of asbestos-induced IPF, MMs are recruited to fibrotic regions in response to inflammatory and chemotactic signals. Within these regions, they exert pro-fibrotic effects by phagocytosing asbestos fibers, localizing to pro-fibrotic niches, and facilitating interactions between injured alveolar epithelial cells and fibroblasts. Overexpression of the SIRT3 gene ameliorates the progression of IPF by reducing the recruitment of Mo-AMs [109]. Importantly, Mo-AMs can enhance fibroblast activation and proliferation through the release of platelet-derived growth factor A (PDGF-A) and sustain their own self-renewal via the secretion of M-CSF, thereby perpetuating a vicious cycle (Table 1). Exposure of lung tissue to bleomycin results in significant pulmonary inflammation, which can lead to alveolar epithelial cell damage. This damage further reduces the population of tissue-resident alveolar macrophages and facilitates the recruitment of circulating monocytes. Upon entry into the tissue, these monocytes differentiate into Mo-AMs. The sustained presence of Mo-AMs results in the secretion of substantial amounts of pro-fibrotic factors, thereby contributing to the pathogenesis of idiopathic pulmonary fibrosis. Research has demonstrated that regulatory T cells (Tregs) play a crucial role in supplying IL-10 during lung injury, which promotes the maturation of Mo-AMs into a tissue-resident phenotype and mitigates the progression of fibrosis. However, in the absence of inflammation, IL-10 may induce Mo-AMs to enter an accelerated repair state, potentially exacerbating pulmonary fibrosis [110]. Heparin-binding epidermal growth factor-like growth factor (HB-EGF), a significant ligand of the epidermal growth factor receptor released by alveolar epithelial cells, monocytes, and macrophages, has been shown to reduce the migration of monocytes and fibroblasts in a CCL2-dependent manner [111]. Upon infiltration into lung tissue, monocytes initially differentiate into Mo-IMs before subsequently transforming into Mo-AMs. Mo-AMs are widely recognized as pivotal effector cells in the pathophysiology of pulmonary fibrosis (PF), as they secrete substantial quantities of pro-fibrotic cytokines and actively participate in the remodeling of the extracellular matrix, thereby facilitating fibrotic processes. The depletion of Mo-AMs and the consequent reduction in the release of pro-fibrotic mediators have been demonstrated to mitigate pulmonary fibrosis. Recent studies have shown that mannosylated albumin nanoparticles incorporating TGFβ1 small-interfering RNA (MANPs) can be internalized by $CD206^{+}$ MMs within lung tissue, leading to a significant reduction in their secretion of pro-fibrotic cytokines and an amelioration of pulmonary fibrosis [112]. Furthermore, some research suggests that Mo-IMs and fibroblasts engage in interactions mediated by signaling molecules such as transforming growth factor-β (TGF-β), secreted phosphoprotein 1 (SPP1), and platelet-derived growth factor (PDGF), which collectively contribute to the progression of fibrosis [113]. In the context of pulmonary fibrosis, there is a significant elevation in the levels of IM-1, IM-3, and IM-4 within interstitial macrophages. Notably, IM-1 demonstrates a pro-fibrotic phenotype, contributing to extracellular matrix remodeling through mechanisms such as Mmp12 and Mmp14 activity, and promoting the secretion of pro-fibrotic cytokines, including Tgfb1 and Pdgfb. Conversely, IM-3 plays a role in modulating the expression of genes related to antigen presentation, thereby influencing the pulmonary immune microenvironment. Meanwhile, IM-4 is characterized by the high expression of genes linked

Table 1 Regulatory pathways of monocyte recruitment in idiopathic pulmonary fibrosis

Year	Authors	Genes	Subjects	Model induced	Results	References
2004	Okuma T, et al	CCR2	C57BL/6 mice	Bleomycin (4 mg/kg)	The mutual recognition of CCL2 and CCR2 induces the recruitment of macrophages and circulating monocytes to sites of lung injury, accompanied by increased expression of MMP-2 and MMP-9, thereby promoting the progression of IPF	[143]
2015	Pilling D, et al	Lumican	C57BL/6 mice	Bleomycin (3 U/kg)	Inflammatory signaling stimulates fibroblasts to secrete Lumican, which subsequently promotes the differentiation of monocyte-derived fibroblasts and the subsequent release of various cytokines (such as IL-13, TGF-β, CTGF, and TNF-α) and extracellular matrix components, thereby facilitating the progression of IPF	[122]
2015	Shichino S, et al	CCR2	C57BL/6 mice	Crystalline silica particles (400 mg/kg) or bleomycin (1.25 mg/kg)	Silica induces circulating monocytes to migrate into lung tissue in a CCR2-dependent manner and differentiate into macrophages. These cells promote pulmonary inflammation via the secretion of pro-inflammatory mediators and MMPs, disrupt lung structure and function, and mediate the progression of bleomycin-induced IPF	[144]
2016	Liu H, et al	MCPIP1	The human monocytic cell line U937	SiO_2 (50 μg/cm^2)	MCPIP1 promotes the progression of IPF by activating p53-mediated macrophage autophagy, apoptosis, and activation, thereby further regulating the proliferation and migration of lung fibroblasts	[101]
2018	McCubbrey AL, et al	CD11b	C57BL/6 mice	Bleomycin (3U/kg)	CD11b^{hi} MMs release substantial amounts of the pro-fibrotic chemokines CCL2 and CCL24	[108]
2019	Bellamri N, et al	CSF1	Human peripheral blood mononuclear cells	20 ng/mL LPS + 20 ng/mL IFN or 20 ng/mL IL-4 + 20 ng/mL IL-13	Nintedanib inhibits the kinase activity of CSF1/CSF1R and suppresses CCL2 expression via the PI3K–Akt pathway, thereby reducing monocyte recruitment and exerting anti-pulmonary fibrotic effects	[136]
2020	Cui H, et al	ApoE	C57BL/6 mice	Bleomycin (1.5U/kg)	ApoE interacts with LRP1 and facilitates the binding of ApoE to collagen I, promoting collagen phagocytosis and further accelerating the progression of IPF	[116]
2020	Zhang N, et al	Notch	C57BL/6 mice	Bleomycin (2.5 mg/kg)	Upon activation, Notch translocates to the nucleus by binding to the recombination signal binding protein Jκ (RBPJκ), leading to increased accumulation of inflammatory factors, TGF-β, and collagen. This process promotes the recruitment of monocyte-derived Ly6C^{lo}MHCII^{hi} macrophages and drives the progression of IPF	[117]
2020	Joshi N, et al	M-CSFR	C57BL/6 mice	TiO_2 (100 μg/ mouse) or asbestos fiber (100 μg/mouse)	Mo-AMs recruit monocytes and sustain their own proliferation via M-CSF/M-CSFR signaling, thereby mediating the advancement of IPF	[145]
2021	Venosa A, et al	CCR2	C57BL/6 mice	SP-C^{I73T}	Ablation of CCR2$^+$ bone marrow-derived monocyte subsets effectively suppresses the continued deterioration of the pulmonary fibrotic phenotype	[146]
2021	Cheresh P, et al	SIRT3	C57BL/6 mice	Containing TiO_2 or asbestos fiber suspension (100 μg/mouse)	Overexpression of the SIRT3 gene ameliorates the progression of IPF by reducing the recruitment of Mo-AMss	[109]

Table 1 (continued)

Year	Authors	Genes	Subjects	Model induced	Results	References
2021	Brody SL, et al	CCR2	C57BL/6 mice	Bleomycin (3 U/kg)	The numbers of $CCR2^+$ monocytes and IMs are significantly increased. These cells can release TGF-β1, which induces fibroblast proliferation and activation as well as excessive deposition of ECM	[147]
2021	Chen B, et al	GM-CSF/ MCP 1	SD rats	TAA (200 mg/kg)	In a rat model of hepatopulmonary syndrome, neutralization of GM-CSF and inhibition of MCP-1 act on bone marrow-derived macrophages, reducing macrophage recruitment and differentiation, suppressing the release of pro-fibrotic factors such as TGF-β1, and thereby ameliorating IPF	[148]
2022	To S, et al	CDH11	C57BL/6 mice	Bleomycin (35 U/kg, intraperitoneal injection)	In CDH11 gene-deficient mice, the number of MMs is markedly reduced, which further suppresses the populations of IMs and AMs in the lung tissue, leading to a sharp decline in macrophage polarization levels	[115]
2022	Bu N, et al	miR-29a-3p	C57BL/6 mice	Nd_2O_3 (25 mg/mouse)	Promoting high expression of lncRNA H19 in macrophages downregulates the expression of TNFRSF1A and miR-29a-3p, activates the NF-κB pathway, and enhances the release of TNF-α, MIP-1α, and IL-6, thereby exacerbating IPF	[121]
2023	Zhao W, et al	MMP19	C57BL/6 mice	Bleomycin (1.5 U/kg)	MMP19 stimulates monocyte infiltration into lung tissue by modulating the SDF1/CXCR4 pathway, thereby exacerbating the progression of IPF	[107]
2023	Zhang G, et al	CCL7	C57BL/6 mice	Bleomycin (3 U/kg)	AnSC-Exos are enriched with let-7b and let-7a miRNAs, which inhibit CCL7 expression in fibroblasts, reducing the recruitment of macrophages and circulating MMs. The decreased recruitment of M2 macrophages contributes to the alleviation of IPF	[139]
2023	Choi SM, et al	CCL2/CCR2	C57BL/6 mice	Bleomycin (3 mg/kg)	Mesenchymal stem cells suppress monocyte infiltration and macrophage differentiation via the CCL2/CCR2 pathway, leading to improved lung function in a mouse model of IPF	[140]
2023	Hong SY, et al	SHP-1	C57BL/6 mice	Bleomycin (3.5 mg/kg)	Inhibition of CSF1R expression in macrophages suppresses downstream STAT3/ NFκB signaling, concurrently inhibiting both M1 and M2 macrophage polarization and blocking the transformation of fibroblasts into myofibroblasts	[118]
2024	Farooq H, et al	CCR2	C57BL/6 mice	Bleomycin (1.6 U/kg) and silica suspension (20 mg/kg)	CCR2 responds to chemokine signals such as CCL2 in lung tissue, facilitating its migration to injured pulmonary sites and differentiation into pro-fibrotic IMs. These cells produce substantial amounts of TGF-β1, thereby promoting the progression of IPF	[149]
2024	Moos PJ, et al	SP-C	C57BL/6 mice	Tamoxifen (90 mg/kg)	SP-C is involved in all stages of monocyte-mediated inflammatory exacerbation and enhances the regulatory role of monocytes over pro-fibrotic signals including SPP1, FN, and TGF-β1	[150]
2024	Chen Y, et al	NAMPT	C57BL/6 mice	Bleomycin (3 mg/kg)	CCR5 and TLR4 effectively activate NAMPT/ STAT6-induced M2 polarization of macrophages and significantly exacerbate lung fibrotic injury	[41]

Table 1 (continued)

Year	Authors	Genes	Subjects	Model induced	Results	References
2025	Cui H, et al	TREM2	C57BL/6 mice	Bleomycin (1.5 U/kg)	TREM2, highly expressed in Mo-AMs, promotes IPF progression through the release of abundant pro-fibrotic mediators such as TGF-β, PDGF-α, osteopontin (OPN), and MMPs	[119]
2025	Cheng PP, et al	Midkine	C57BL/6 mice	aerosols containing PM2.5 or carbon black (1 h/day)	Exposure to PM2.5 induces increased expression of midkine, which triggers substantial release of MCP-1 via the Syndecan-4 receptor on pleural mesothelial cells, thereby promoting the recruitment of circulating monocytes and accumulation of macrophages	[120]

to cell migration, which aids in the recruitment and accumulation of immune cells. The activation of Yes-associated protein (YAP) and transcriptional coactivator with PDZ-binding motif (TAZ) in lung tissue macrophages is observed in bleomycin-induced idiopathic pulmonary fibrosis [114]. This activation is intricately linked to the recruitment of Mo-AMs and damage to alveolar epithelial cells. Targeted knockout of YAP/TAZ markedly diminishes Mo-AM recruitment, mitigates inflammatory and pro-fibrotic responses, and promotes the regeneration of alveolar epithelial cells.

Regulation of monocyte recruitment in IPF—other pathways

Beyond the classical chemoattractant pathways, the recruitment of monocytes to injured lung tissue is modulated by a variety of factors. In a BLM-induced mouse model of IPF, there was a significant upregulation of CDH11 expression. Intervention with a CDH11-neutralizing antibody or the use of CDH11-deficient mice resulted in a marked reduction in the extent of fibrosis [115]. Ex vivo analyses revealed that bone marrow cells lacking CDH11 exhibited a diminished capacity for macrophage differentiation when stimulated with 20 ng/mL M-CSF. In CDH11-deficient mice, there was no significant increase in M2 macrophages, a phenomenon correlated with a substantial reduction in MMs. This reduction in monocytes may further decrease the populations of both interstitial and AMs in lung tissue, thereby leading to a pronounced decline in macrophage polarization levels. Transcriptomic analyses indicate significant differences between TRMs and MMs. Notably, Mo-AMs are the primary source of Apolipoprotein E (ApoE), a crucial molecule involved in lipid metabolism, within lung tissue. ApoE interacts with the low-density lipoprotein receptor-related protein 1 (LRP1), facilitating the binding of ApoE to collagen I, which enhances collagen phagocytosis and thereby contributes to the progression of IPF [116]. The Notch signaling pathway plays a pivotal role in regulating macrophage activation and polarization. Upon activation, Notch translocates to the nucleus by binding to the recombination signal binding protein for immunoglobulin kappa J region (RBP-J), thereby modulating downstream gene expression [117]. Genetic knockout of RBP-J not only effectively suppresses the expression of the Notch signaling pathway but also reduces the accumulation of inflammatory factors, transforming growth factor-beta (TGF-β), and collagen. Furthermore, it inhibits the recruitment of monocyte-derived $Ly6C^{lo}MHCII^{hi}$ macrophages, thereby mitigating the progression of IPF. In addition, during renal fibrosis in mice, the Notch signaling pathway in macrophages promotes fibrosis progression by enhancing the infiltration and activation of circulating MMs. Inhibition of the Notch signaling pathway within the bone marrow has been shown to effectively suppress the recruitment of monocyte-derived $Ly6C^{lo}MHCII^{hi}$ macrophages. This suppression subsequently reduces the secretion of pro-fibrotic mediators, such as TGF-β1, and diminishes the activation of myofibroblasts, thereby exerting an anti-fibrotic effect in the lungs. Targeting macrophages, or specific subsets of MMs, may therefore offer therapeutic benefits for the treatment of IPF. Both MMs and AMs contribute to pulmonary inflammation through the secretion of pro-inflammatory mediators and matrix metalloproteinases (MMPs), which can disrupt lung structure and function and facilitate the progression of BLM-induced IPF. Src homology domain 2 (SH2)-containing tyrosine phosphatase-1 (SHP-1), a protein tyrosine phosphatase, plays a significant role in the progression of IPF by inhibiting the hyperactivation of innate immune cells, including AMs [118]. Research indicates that SHP-1 deficiency results in a substantial accumulation of macrophages within lung tissue, thereby increasing susceptibility to IPF. Recent studies suggest that Mo-AMs, as opposed to tissue-resident AMs, facilitate the progression of IPF by secreting substantial amounts of pro-fibrotic mediators, including TGF-β1, PDGF-α, osteopontin (OPN), and MMPs. TREM2, a member of the TREM family, is markedly expressed in Mo-AMs in the lungs of mice afflicted with IPF and contributes to macrophage proliferation and differentiation towards a pro-fibrotic phenotype [119]. Notably, the absence or inhibition of TREM2 significantly mitigated BLM-induced IPF in murine models.

Prolonged exposure to PM2.5 significantly enhances the expression of midkine, a heparin-binding growth factor that facilitates the migration of neurons, neutrophils, and macrophages. This upregulation results in the substantial release of monocyte chemoattractant protein-1 (MCP-1) through the receptor Syndecan-4 (SDC4) on pleural mesothelial cells (PMCs), thereby facilitating the recruitment of circulating monocytes and the accumulation of macrophages [120]. Concurrent exposure to PM2.5 and BLM, which induces elevated expression of CCR2 in monocytes and macrophages, fosters M1 macrophage polarization and the secretion of pro-fibrotic inflammatory mediators. This process ultimately drives various lung tissue cell types, including PMCs, epithelial cells, and fibroblasts, towards a fibrotic phenotype. Importantly, M1 macrophages secrete both pro-inflammatory cytokines and pro-fibrotic factors. Serum amyloid P (SAP) plays a crucial role in the regulation of innate immunity by inhibiting the differentiation of monocytes into fibroblasts and reducing the accumulation of neutrophils and ECM at sites of injury, while simultaneously promoting the formation of regulatory macrophages. SAP plays a crucial role in various fibrotic diseases. Research indicates that modulating SAP can suppress fibrosis in several animal models. Clinical trials have demonstrated that intravenous administration of SAP can reverse the decline in lung function in patients with IPF [115]. In a rat model of lung fibrosis induced by neodymium oxide (Nd_2O_3), it has been observed that Nd_2O_3 promotes the high expression of lncRNA H19 in macrophages. This expression downregulates TNFRSF1A and targets miR-29a-3p, a key factor in lung injury associated with IPF, which reduces collagen type I alpha 1 levels. Consequently, this activates the NF-κB pathway and enhances the release of TNF-α, MIP-1α, and IL-6, ultimately promoting IPF [121]. Lumican, a small leucine-rich proteoglycan, is typically expressed at low levels in normal human lung tissue but is significantly upregulated in patients with IPF [122]. Following tissue injury, inflammatory signals such as TNF-α stimulate fibroblasts to secrete Lumican. This secretion promotes fibrocyte differentiation from monocyte-derived fibroblasts and the subsequent release of various cytokines, including IL-13, TGF-β, CTGF, and TNF-α, as well as ECM components, thereby accelerating the progression of IPF. In the context of carbon tetrachloride-induced liver fibrosis, the genetic knockout of Lumican has been shown to significantly attenuate collagen cross-linking and the severity of fibrosis. The SHP-1 agonist, SC-43, demonstrates efficacy in mitigating alveolar hemorrhage, pulmonary inflammation, and collagen deposition within a murine model of IPF. This is accompanied by a notable reduction in macrophage and monocyte populations within lung tissue. During the progression of IPF, macrophages undergo differentiation into M1 and M2 subtypes. An excessive polarization towards the M1 phenotype results in epithelial damage and can trigger an adaptive M2 polarization, which subsequently facilitates fibrotic remodeling. The differentiation of M1 and M2 macrophages is mediated by NF-κB and STAT3 pathways, respectively. SC-43 effectively inhibits the polarization of both M1 and M2 macrophages by suppressing the expression of CSF1R and the downstream STAT3/NF-κB signaling pathways in macrophages, thereby impeding the transition from fibroblasts to myofibroblasts. It is important to note that conditioned medium derived from M2 macrophages effectively inhibits the transformation of fibroblasts into myofibroblasts, whereas conditioned medium from M1 macrophages also suppresses the production of ECM components, including collagen. While Mo-AMs are extensively acknowledged for their positive correlation with the progression of IPF, emerging evidence suggests that Mo-IMs also play a pro-fibrotic role. These IMs are known to upregulate MMP12 and MMP14, which are implicated in ECM remodeling and wound healing, as well as pro-fibrotic cytokines such as TGF-β1 and PDGF-β. Furthermore, macrophages facilitate fibroblast activation and proliferation through SPP1, which enhances fibroblast migration, adhesion, and survival, and PDGF, which stimulates fibroblast proliferation and ECM production, thereby contributing to the advancement of IPF. Nicotinamide phosphoribosyltransferase (NAMPT) is a highly conserved enzyme across species, with its extracellular form (eNAMPT) influencing macrophage polarization and engaging in various inflammatory processes [41]. Recent studies have demonstrated that the conditional knockout of NAMPT in monocytes/macrophages impedes M2 polarization and significantly mitigates pulmonary fibrotic injury. Similarly, the inhibition of C–C chemokine receptor type 5 (CCR5) and Toll-like receptor 4 (TLR4) reduces NAMPT/STAT6-mediated M2 macrophage polarization, likely by suppressing chemokine release and monocyte recruitment.

Classical monocytes, which represent the predominant monocyte subset, have the capacity to migrate to injured lung tissue and differentiate into tissue macrophages. Although most $CCR2^+$ cells are generally considered pro-fibrotic, the $CCR2^+CD4^+$ T cell population may have a role in attenuating fibrosis. Monocytes not only differentiate into macrophages but also engage in interactions with various other cell types. For instance, type 2 innate lymphoid cells (ILC2s) interact with monocytes, thereby further enhancing the fibrotic microenvironment and accelerating the progression of idiopathic pulmonary fibrosis (IPF) [123]. ILC2s play a pivotal role in the activation of monocytes and their subsequent differentiation into alternatively activated macrophages that secrete growth factors. In the context of IPF, ILC2s produce substantial quantities of cytokines, including interleukin-4 (IL-4), interleukin-13 (IL-13), and

interleukin-5 (IL-5), which are instrumental in the activation of monocytes and their differentiation into growth factor-secreting alternatively activated macrophages. Importantly, in mouse models of IPF induced by silica and BLM, the depletion of monocytes or the reduction of their recruitment and activation within lung tissue markedly enhanced lung function. During the initial stages of IPF, monocyte/macrophage-derived chemokines CXCL1 and CXCL2 facilitate the recruitment of neutrophils to the lung. These neutrophils, in turn, release mediators that activate fibroblasts, thereby contributing to the progression of IPF.

Monocyte-associated apoptosis in idiopathic pulmonary fibrosis

In the context of IPF, monocytes recruited to local tissues engage with pulmonary parenchymal cells to modulate apoptotic pathways, thereby impacting the progression of the disease. Apoptosis and the subsequent clearance of apoptotic cells serve as essential regulatory mechanisms in the modulation of inflammation and tissue repair during the fibrotic process, representing a critical aspect of fibrosis. Elevated levels of apoptosis are evident across nearly all forms of fibrosis. Apoptotic cells release substantial amounts of ATP, which is subsequently converted to adenosine. Through adenosine receptors, adenosine inhibits the secretion of pro-inflammatory mediators and chemokines. Macrophages, upon phagocytosing apoptotic cells, release significant quantities of pro-fibrotic factors such as IL-10 and TGF-β1 [124]. It is widely recognized that the interaction with apoptotic cells directs macrophages towards an M2 phenotype polarization. Nevertheless, some studies have indicated that macrophage phagocytosis of apoptotic cells can lead to a reduction in the expression of type I collagen and α-SMA in fibroblasts. Further investigation is warranted to elucidate the interactions between apoptotic cells and macrophages.

The apoptosis of alveolar epithelial cells (AECs) plays a regulatory role in the local recruitment of monocytes and macrophages. In the context of IPF, apoptotic alveolar epithelial type 2 cells (AEC-IIs) release significant quantities of oxidized phospholipids (oxPLs). The CD36 receptor on the macrophage surface facilitates the uptake of these oxPLs, leading to phenotypic alterations in macrophages that enhance the secretion of the pro-fibrotic mediator TGF-β, thereby promoting IPF [125]. Lysosomal phospholipase A2 (LPLA2), a phospholipase markedly upregulated in alveolar macrophages, is a crucial enzyme in the degradation of oxPLs within alveolar tissues. The overexpression of LPLA2 significantly mitigates lipid accumulation in alveolar cells and subsequently reduces both TGF-β release from macrophages and AEC-II apoptosis, ultimately alleviating IPF [126]. In the bleomycin (BLM)-induced IPF model, there is a notable upregulation of RNA-binding protein 7 (RBM7) within lung tissue, which facilitates the degradation of nuclear-enriched abundant transcript 1 (NEAT1). This process promotes apoptosis in alveolar epithelial cells (AECs) and disrupts the repair and regeneration of normal epithelial cells. Consequently, the release of C–X–C motif chemokine ligand 12 (CXCL12) significantly enhances the recruitment of circulating monocytes to the affected tissues, thereby exacerbating IPF [127]. In the context of asbestos-induced IPF, the expression of SIRT3, a mitochondrial protein involved in the regulation of cell fate and senescence, is notably suppressed in AECs. The overexpression of SIRT3 has been shown to effectively mitigate mitochondrial DNA (mtDNA) damage and apoptosis in AECs, leading to a reduction in the local recruitment of MMs and an alleviation of IPF [109].

The apoptotic mechanisms of both MMs and tissue-resident macrophages play a pivotal role in modulating the progression of IPF. In the context of bleomycin (BLM)-induced IPF, microRNAs (miRNAs), which are evolutionarily conserved small non-coding RNAs, are closely linked to disease progression. Specifically, let-7c, miR-125b, and miR-155 are markedly upregulated in this pathological condition. MMs demonstrate pro-fibrotic effects in BLM-induced IPF. A deficiency in miRNAs triggers apoptosis in tissue-resident macrophages within the lung, yet simultaneously recruits MMs, thereby enhancing fibroblast activation and exacerbating IPF [128]. In asbestos-induced IPF, NADPH oxidase 4 (NOX4) leads to a significant increase in both mitochondrial number and mitochondrial reactive oxygen species (ROS) within macrophages, conferring resistance to apoptosis. The conditional knockout of NOX4 in MMs significantly augments macrophage apoptosis and mitigates IPF [129]. In the silica-induced IPF model, silica exposure leads to the polarization of macrophages towards both M1 and M2 phenotypes, characterized by significant upregulation of markers such as inducible nitric oxide synthase (iNOS) and Arginase 1, among others. During this process, AMs modulate p53 signaling via MCPIP1 to induce autophagy, which subsequently results in macrophage activation and apoptosis [101]. Related ex vivo studies demonstrate that conditioned media from silica-exposed macrophages significantly enhance fibroblast proliferation and migration, thereby promoting IPF. Carbon nanotubes, which are extensively utilized as industrial nanomaterials, pose an increased risk of lung tissue exposure upon inhalation. Multi-walled carbon nanotubes that penetrate lung tissue can activate the NLRP3 inflammasome in macrophages, resulting in the release of numerous proteins and the activation of the innate immune response. Consequently, macrophages recruit additional immune cells through the secretion

of signaling molecules and antigen presentation. Notably, PPIA functions as a chemoattractant, recruiting and activating MMs, which initiates an inflammatory cascade. Additionally, macrophages secrete caspase-1, thereby triggering apoptosis. Research has demonstrated a positive correlation between macrophage apoptosis and silica-induced silicosis [130]. Notably, the fibrotic effector TGF-β1 can diminish the antiviral capacity and pro-inflammatory responses of macrophages. This effect is primarily due to its inhibition of the mitochondrial respiratory chain and disruption of mitochondrial function in macrophages, which in turn promotes mitochondrial stress and apoptosis [131]. MMs, as opposed to inactivated tissue-resident macrophages, play a crucial role in host defense. TREM2, a member of the triggering receptor expressed on myeloid cells (TREM) family, is significantly expressed in MMs within the lung tissues of mice with IPF. It facilitates the anti-apoptotic effects of lipid ligands, thereby promoting macrophage survival and differentiation towards a pro-fibrotic phenotype [119]. The apoptotic processes of both monocyte-derived and tissue-resident macrophages have a direct impact on the release of pro-inflammatory and pro-fibrotic factors in lung tissues. Consequently, regulating macrophage apoptosis may offer a promising therapeutic strategy for mitigating the progression of IPF.

Moreover, in addition to IPF, monocyte-associated apoptosis in various tissues contributes significantly to the regulation of fibrotic processes. Chronic muscle injury initiates an innate immune response, wherein myogenic cells and tissue-resident macrophages release a plethora of chemokines that facilitate the recruitment of circulating monocytes. These infiltrating inflammatory macrophages express tumor necrosis factor (TNF), which induces apoptosis in both macrophages and fibro/adipogenic progenitors (FAPs), thereby preventing FAP differentiation into interstitial cells. However, in the context of chronic muscle injury, macrophages exhibit elevated expression of TGF-β1, which inhibits the apoptosis of FAPs and subsequently promotes muscle fibrosis [132]. Acute myocardial infarction, a leading cause of heart failure and mortality, is further complicated by reperfusion injury, which can lead to myocardial remodeling with fibrosis as a central component. The initial phase (days 1–3) of myocardial infarction is marked by significant local accumulation of immune cells, including macrophages and neutrophils. The sustained pro-inflammatory response results in the secretion of substantial amounts of pro-inflammatory mediators, such as interleukin-1β (IL-1β), interleukin-6 (IL-6), and tumor necrosis factor-alpha (TNF-α), as well as pro-fibrotic mediators, including transforming growth factor-beta (TGF-β). This cascade subsequently triggers fibroblast activation, thereby facilitating the fibrotic process and cardiac remodeling. Empirical evidence suggests that the inhibition of CD11b-targeted monocyte/macrophage infiltration into local tissues significantly diminishes cardiac cell apoptosis and reduces the extent of myocardial infarction [133]. In the context of hepatic inflammation, there is a notable upregulation of Interferon-γ-inducible protein 10 (IP-10) expression in monocytes. IP-10 has the capacity to directly induce hepatocyte apoptosis through the CXCR3 receptor, and its pronounced elevation serves as a critical biomarker of fibrosis [134]. Oils, predominantly comprising mineral oil, peanut oil, corn oil, and olive oil, are frequently utilized as solvents for the administration of lipophilic substances into the peritoneal cavity of experimental animals. In models of liver fibrosis, the intraperitoneal injection of carbon tetrachloride, with vegetable oil serving as the solvent, has been observed to exacerbate liver fibrosis. Within the peritoneal environment, vegetable oil can be internalized by macrophages, thereby activating the caspase-3 cascade, which results in foam cell formation and apoptosis. This apoptotic process subsequently recruits circulating monocytes from the bloodstream to the peritoneum, where they differentiate into macrophages, thereby facilitating the formation of inflammatory granulomas and the progression of fibrosis [135].

Monocyte-based anti-idiopathic pulmonary fibrosis therapy

Colony-stimulating factor 1 (CSF1), the principal hematopoietic growth factor for monocytes, interacts with its receptor (CSF1R) to facilitate the recruitment of monocytes to damaged tissues and their subsequent differentiation into mature macrophages [136]. Additionally, CSF1 stimulates macrophages to secrete significant quantities of chemoattractants, such as CCL2, which further enhances the accumulation of monocytes at sites of lung injury, thereby contributing to the development of IPF. Nintedanib, a drug approved for its anti-fibrotic properties, has been shown to effectively inhibit the kinase activity of CSF1R. It also downregulates both mRNA and protein expression of CCL2 via the PI3K-Akt signaling pathway, thereby diminishing monocyte recruitment and exerting anti- IPF effects [137, 138]. Furthermore, Nintedanib limits the activation of mast cells and specific myeloid cell populations. It reduces the recruitment and activation of circulating monocytes to injured lung tissue by inhibiting the macrophage colony-stimulating factor 1 receptor (CSF1R) in bone marrow-derived macrophages and decreasing CCL2 levels. By impairing the accumulation of $CCR2^{+}$ cells in lung tissue, Nintedanib effectively suppresses the progression of IPF [136]. Exogenous antigens promote the polarization of macrophages towards the M2 phenotype through the injury of alveolar epithelial cells. The subsequent release of TGF-β1

facilitates the differentiation of fibroblasts into myofibroblasts, which are the primary effector cells in IPF, via the TGF-β1/Smad signaling pathway, resulting in excessive production of ECM. MicroRNAs (miRNAs) are among the most critical molecular constituents of exosomes. Exosomes derived from Antler Stem Cells (AnSC-Exos) are particularly enriched with let-7b and let-7a miRNAs. These miRNAs suppress the expression of CCL7, a monocyte chemoattractant, in fibroblasts, thereby reducing the recruitment of macrophages and circulating MMs. This reduction in M2 macrophage recruitment contributes to the mitigation of IPF [139]. Ly6C is utilized as a marker to distinguish classical monocytes, which can be categorized into classical and non-classical phenotypes. In the context of BLM-induced IPF, there is a notable increase in the population of classical monocytes. Mesenchymal stem cells (MSCs), a type of stromal cell with stem cell properties, possess the capacity to proliferate and differentiate into various cell types under specific conditions. In the context of IPF, the intravenous administration of mesenchymal stromal/stem cells (MSCs) has been shown to effectively reduce the number of classical MMs and the content of M2 macrophages within lung tissue, thereby enhancing lung function in murine models of the disease [140]. MSCs exhibit significant therapeutic potential against idiopathic IPF. Extracellular vesicles and exosomes derived from MSCs (MEx) have been demonstrated to effectively decrease the proportions of "pro-inflammatory" classical monocytes, non-classical monocytes, and AMs in the lungs. This modulation of lung macrophage phenotypes contributes to the prevention of BLM-induced IPF progression [141]. Furthermore, the transplantation of bone marrow-derived macrophages pretreated with MEx from lung, liver, and spleen sources has also been shown to provide protective effects against BLM-induced lung fibrosis. The anti-inflammatory prostaglandin 15d-PGJ2 is primarily synthesized from arachidonic acid. This synthesis involves the conversion of arachidonic acid to prostaglandin H2 via cyclooxygenase, followed by its transformation to prostaglandin D2 by hematopoietic prostaglandin D synthase, and subsequent dehydration [142]. 15d-PGJ2 is capable of modulating hepatocyte growth factor (HGF) through autocrine or paracrine mechanisms. This modulation effectively inhibits the FMT and EMT induced by TGF-β1, thereby exhibiting significant anti-fibrotic properties. Although curcumin is characterized by low oral bioavailability, its oral administration, as opposed to intravenous injection, has been shown to exert notable anti-fibrotic effects in models of IPF induced by BLM and silica. Curcumin facilitates the intestinal synthesis of 15d-PGJ2, which subsequently activates the PPARγ and CREB signaling pathways, leading to enhanced HGF release. The HGF then circulates to the lung tissue via the bloodstream, where it mediates its anti-fibrotic effects.

Summary and prospects

Monocytes and macrophages, integral components of the innate immune system, are crucial for maintaining physiological homeostasis. In addition to their role as immune defenders, TRMs are involved in tissue development and remodeling, with their specific characteristics and functions being influenced by their ontogeny and the local microenvironment. Although monocytes and macrophages are part of the mononuclear phagocyte system, they originate from distinct developmental pathways and exhibit unique functional properties. Upon entering new tissue environments, MMs adopt functions akin to those of resident macrophage populations, contributing to either the initiation or resolution of inflammation, contingent upon the experimental conditions and the context of their differentiation. Notably, MMs exhibit significant similarities to pro-fibrotic macrophages present in the lung tissues of individuals with IPF, sharing common pro-fibrotic signaling pathways. The heterogeneity of macrophages presents substantial implications for the prospective treatment of pulmonary fibrotic diseases. Targeting pro-fibrotic macrophages with precision, as opposed to broadly targeting M1 (pro-inflammatory) or M2 (anti-inflammatory) macrophage phenotypes, may more effectively meet clinical requirements. Monocytes are generated through embryonic and adult hematopoiesis in the fetal liver and bone marrow, respectively. Upon recruitment to sites of tissue injury, monocytes have the potential to differentiate into macrophages and dendritic cells; however, the differentiation tendencies of various monocyte subsets across different tissues remain insufficiently explored. Macrophages and dendritic cells exhibit diverse functional roles across a spectrum of diseases. Therefore, the precise modulation of the recruitment and differentiation of specific monocyte subsets, tailored to their unique functions, rather than the broad inhibition of monocyte recruitment and differentiation in injured tissues, is more likely to yield therapeutic benefits.

Macrophages within lung tissue are essential cells that secrete inflammatory mediators and chemoattractants. Circulating monocytes, once recruited to the lung, have the potential to differentiate into MMs. The extent to which these macrophages contribute to the release of significant quantities of chemoattractants and inflammatory mediators, thereby facilitating the persistent progression of IPF and potentially creating a self-perpetuating cycle, remains to be elucidated. Current investigations into IPF predominantly employ murine or rat models; however, the applicability

of these findings to human IPF has yet to be conclusively established. Animal models of IPF are typically induced using agents such as BLM and silica, but these models fail to fully replicate the pathological mechanisms observed in human IPF. Future research should incorporate human samples to elucidate the critical role of monocytes in the pathogenesis of IPF. Additionally, the majority of IPF models are developed using male mice, and the potential existence of sex-based differences in the regulation of monocytes during the development of IPF requires further investigation. While silica and BLM serve as models for IPF induced by various factors, it is imperative to establish standardized criteria—encompassing the dosage of modeling agents and the associated phenotypic characteristics—to enable a thorough investigation into the pivotal roles of monocytes in IPF caused by diverse factors. Monocytes can be classified into classical, intermediate, and non-classical subsets. Current research predominantly focuses on inhibiting monocyte recruitment to the lungs; however, further clarification is needed regarding which subset is more predisposed to pulmonary recruitment. This understanding is crucial for developing targeted interventions for specific monocyte subtypes. Importantly, existing studies on the role of monocytes in IPF are predominantly limited to the early stages, potentially linked to the reduction in the origin and number of lung tissue macrophages mediated by monocytes. Nonetheless, the functions of monocytes across different stages of IPF warrant comprehensive examination to establish a foundation for future research directions and in-depth investigations.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Koudstaal T, Funke-Chambour M, Kreuter M, Molyneaux PL, Wijsenbeek MS (2023) Pulmonary fibrosis: from pathogenesis to clinical decision-making. Trends Mol Med 29:1076–1087
2. Thannickal VJ, Toews GB, White ES, Lynch JP, Martinez FJ (2004) Mechanisms of pulmonary fibrosis. Annu Rev Med 55:395–417
3. Yang L, Li Y, Zeng T et al (2025) Global, regional, and national burden of hypertension-induced cardiorenal disease, 1990-2021 and projections to 2050: a systematic analysis for the global burden of disease study 2021. BMC Public Health 25:2596
4. Spagnolo P, Kropski JA, Jones MG et al (2021) Idiopathic pulmonary fibrosis: disease mechanisms and drug development. Pharmacol Ther 222:107798
5. Bonella F, Spagnolo P, Ryerson C (2023) Current and future treatment landscape for idiopathic pulmonary fibrosis. Drugs 83:1581–1593
6. Morse C, Tabib T, Sembrat J et al (2019) Proliferating SPP1/MERTK-expressing macrophages in idiopathic pulmonary fibrosis. Eur Respir J. https://doi.org/10.1183/13993003.02441-2018
7. Su L, Dong Y, Wang Y et al (2021) Potential role of senescent macrophages in radiation-induced pulmonary fibrosis. Cell Death Dis 12:527
8. Zhang M, Zhang J, Hu H et al (2024) Multiomic analysis of monocyte-derived alveolar macrophages in idiopathic pulmonary fibrosis. J Transl Med 22:598
9. Liu L, Wu P, Wei Y et al (2025) TWEAK-Fn14 signaling protects mice from pulmonary fibrosis by inhibiting fibroblast activation and recruiting pro-regenerative macrophages. Cell Rep 44:115220
10. Misharin AV, Morales-Nebreda L, Reyfman PA et al (2017) Monocyte-derived alveolar macrophages drive lung fibrosis and persist in the lung over the life span. J Exp Med 214:2387–2404
11. Yuan H, You Y, He Y et al (2023) Crystalline silica-induced proinflammatory interstitial macrophage recruitment through Notch3 signaling promotes the pathogenesis of silicosis. Environ Sci Technol 57:14502–14514
12. Raghu G, Mouded M, Chambers DC et al (2022) A phase IIb randomized clinical study of an anti-α(v)β(6) monoclonal antibody in idiopathic pulmonary fibrosis. Am J Respir Crit Care Med 206:1128–1139
13. John AE, Graves RH, Pun KT et al (2020) Translational pharmacology of an inhaled small molecule αvβ6 integrin inhibitor for idiopathic pulmonary fibrosis. Nat Commun 11:4659
14. Wang J, Xu L, Xiang Z et al (2020) Microcystin-LR ameliorates pulmonary fibrosis via modulating CD206(+) M2-like macrophage polarization. Cell Death Dis 11:136
15. Spagnolo P, Tonelli R, Samarelli AV et al (2022) The role of immune response in the pathogenesis of idiopathic pulmonary fibrosis: far beyond the Th1/Th2 imbalance. Expert Opin Ther Targets 26:617–631
16. Okamoto M, Kuratani A, Okuzaki D et al (2024) Tff1-expressing Tregs in lung prevent exacerbation of Bleomycin-induced pulmonary fibrosis. Front Immunol 15:1440918
17. Guo L, Karoubi G, Duchesneau P et al (2018) Interrupted reprogramming of alveolar type II cells induces progenitor-like cells that ameliorate pulmonary fibrosis. NPJ Regen Med 3:14
18. Bueno M, Calyeca J, Rojas M, Mora AL (2020) Mitochondria dysfunction and metabolic reprogramming as drivers of idiopathic pulmonary fibrosis. Redox Biol 33:101509
19. Wang F, Ting C, Riemondy KA et al (2023) Regulation of epithelial transitional states in murine and human pulmonary fibrosis. J Clin Investig. https://doi.org/10.1172/JCI165612

20. Wang H, Xu H, Lyu W et al (2022) KLF4 regulates TERT expression in alveolar epithelial cells in pulmonary fibrosis. Cell Death Dis 13:435
21. Mercer PF, Johns RH, Scotton CJ et al (2009) Pulmonary epithelium is a prominent source of proteinase-activated receptor-1-inducible CCL2 in pulmonary fibrosis. Am J Respir Crit Care Med 179:414–425
22. Yang J, Agarwal M, Ling S et al (2020) Diverse injury pathways induce alveolar epithelial cell CCL2/12, which promotes lung fibrosis. Am J Respir Cell Mol Biol 62:622–632
23. Yang W, Pan L, Cheng Y et al (2022) Nintedanib alleviates pulmonary fibrosis in vitro and in vivo by inhibiting the FAK/ERK/S100A4 signalling pathway. Int Immunopharmacol 113:109409
24. Li Y, Bao J, Bian Y et al (2018) S100A4(+) macrophages are necessary for pulmonary fibrosis by activating lung fibroblasts. Front Immunol 9:1776
25. Gao P, Lu Y, Tang K et al (2024) Ficolin-1 ameliorates pulmonary fibrosis via directly binding to TGF-β1. J Transl Med 22:1051
26. Du C, Ma C, Geng R et al (2025) Bruceine A inhibits TGF-β1/Smad pathway in pulmonary fibrosis by blocking gal3/TGF-β1 interaction. Phytomedicine 136:156267
27. Batra R, Suh MK, Carson JS et al (2018) IL-1β (Interleukin-1β) and TNF-α (Tumor Necrosis Factor-α) Impact Abdominal Aortic Aneurysm Formation by Differential Effects on Macrophage Polarization. Arterioscler Thromb Vasc Biol 38:457–463
28. Liu C, Wu K, Li C et al (2024) SPP1+ macrophages promote head and neck squamous cell carcinoma progression by secreting TNF-α and IL-1β. J Exp Clin Cancer Res CR 43:332
29. Cui Y, Chen C, Tang Z et al (2024) TREM2 deficiency aggravates renal injury by promoting macrophage apoptosis and polarization via the JAK-STAT pathway in mice. Cell Death Dis 15:401
30. Cao M, Yan H, Han X et al (2019) Ginseng-derived nanoparticles alter macrophage polarization to inhibit melanoma growth. J Immunother Cancer 7:326
31. Li J, Zeng G, Zhang Z et al (2024) Urban airborne PM(2.5) induces pulmonary fibrosis through triggering glycolysis and subsequent modification of histone lactylation in macrophages. Ecotoxicol Environ Saf 273:116162
32. Liu TT, Sun HF, Tang MZ et al (2024) Bicyclol attenuates pulmonary fibrosis with silicosis via both canonical and non-canonical TGF-β1 signaling pathways. J Transl Med 22:682
33. Yang Y, Xiao Z, Yang W et al (2024) Role of transient receptor potential ankyrin 1 in idiopathic pulmonary fibrosis: modulation of M2 macrophage polarization. Cell Mol Life Sci CMLS 81:187
34. Hiroki CH, Hassanabad MF, Defaye M et al (2025) Nociceptor neurons suppress alveolar macrophage-induced Siglec-F(+) neutrophil-mediated inflammation to protect against pulmonary fibrosis. Immunity 58:2054-2068.e2056
35. Yuan X, Qin X, Takemoto K et al (2025) Human hypofunctional NCF1 variants promote pulmonary fibrosis in the bleomycin-induced mouse model and patients with systemic sclerosis via expansion of SPP1(+) monocytes-derived macrophages. Ann Rheum Dis 84:294–306
36. Suzuki M, Ikari J, Anazawa R et al (2020) PAD4 deficiency improves bleomycin-induced neutrophil extracellular traps and fibrosis in mouse lung. Am J Respir Cell Mol Biol 63:806–818
37. Sun X, Zhang X, He Y, Du X, Cai Q, Liu Z (2025) CD4(+)T and CD8(+)T cells profile in lung inflammation and fibrosis: targets and potential therapeutic drugs. Front Immunol 16:1562892
38. Mutsaers SE, Miles T, Prêle CM, Hoyne GF (2023) Emerging role of immune cells as drivers of pulmonary fibrosis. Pharmacol Ther 252:108562
39. Kumar V, Hertz M, Agro A, Byrne AJ (2023) Type 1 invariant natural killer T cells in chronic inflammation and tissue fibrosis. Front Immunol 14:1260503
40. Yuan H, He Y, Zhang Y, Min H, Chen J, Li C (2024) Crystalline silica-induced endoplasmic reticulum stress promotes the pathogenesis of silicosis by augmenting proinflammatory interstitial pulmonary macrophages. Sci Total Environ 946:174299
41. Chen Y, Wang T, Liang F et al (2024) Nicotinamide phosphoribosyltransferase prompts bleomycin-induced pulmonary fibrosis by driving macrophage M2 polarization in mice. Theranostics 14:2794–2815
42. Doan CA, Sabin FR (1926) Normal and pathological fragmentation of red blood cells; the phagocytosis of these fragments by desquamated endothelial cells of the blood stream; the correlation of the peroxidase reaction with phagocytosis in mononuclear cells. J Exp Med 43:839–850
43. Lu H, Suo Z, Lin J, Cong Y, Liu Z (2024) Monocyte-macrophages modulate intestinal homeostasis in inflammatory bowel disease. Biomark Res 12:76
44. Adams RC, MacDonald KPA, Hill GR (2025) The contribution of the monocyte-macrophage lineage to immunotherapy outcomes. Blood 145:1010–1021
45. Wang Z, Xie L, Ding G et al (2021) Single-cell RNA sequencing of peripheral blood mononuclear cells from acute Kawasaki disease patients. Nat Commun 12:5444
46. Arpinati L, Scherz-Shouval R (2023) From gatekeepers to providers: regulation of immune functions by cancer-associated fibroblasts. Trends Cancer 9:421–443
47. Li C, Ture SK, Nieves-Lopez B et al (2024) Thrombocytopenia independently leads to changes in monocyte immune function. Circ Res 134:970–986
48. Torres-Ruiz J, Carrillo-Vazquez DA, Padilla-Ortiz DM et al (2020) TLR expression in peripheral monocyte subsets of patients with idiopathic inflammatory myopathies: association with clinical and immunological features. J Transl Med 18:125
49. Shi C, Jia T, Mendez-Ferrer S et al (2011) Bone marrow mesenchymal stem and progenitor cells induce monocyte emigration in response to circulating toll-like receptor ligands. Immunity 34:590–601
50. Liner AG, van Gogh M, Roblek M, Heikenwalder M, Borsig L (2025) Non-redundant roles of the CCR1 and CCR2 chemokine axes in monocyte recruitment during lung metastasis. Neoplasia 59:101089
51. Regan DP, Chow L, Das S et al (2022) Losartan blocks osteosarcoma-elicited monocyte recruitment, and combined with the kinase inhibitor Toceranib, exerts significant clinical benefit in canine metastatic osteosarcoma. Clin Cancer Res 28:662–676
52. Kumar R, Nolan K, Kassa B et al (2025) Monocytes and interstitial macrophages contribute to hypoxic pulmonary hypertension. J Clin Invest. https://doi.org/10.1172/JCI176865
53. Bajpai G, Bredemeyer A, Li W et al (2019) Tissue resident CCR2- and CCR2+ cardiac macrophages differentially orchestrate monocyte recruitment and fate specification following myocardial injury. Circ Res 124:263–278
54. Jakubzick CV, Randolph GJ, Henson PM (2017) Monocyte differentiation and antigen-presenting functions. Nat Rev Immunol 17:349–362
55. Biswas SK, Mantovani A (2010) Macrophage plasticity and interaction with lymphocyte subsets: cancer as a paradigm. Nat Immunol 11:889–896
56. Barrett TJ, Corr EM, van Solingen C et al (2021) Chronic stress primes innate immune responses in mice and humans. Cell Rep 36:109595
57. Geissmann F, Manz MG, Jung S, Sieweke MH, Merad M, Ley K (2010) Development of monocytes, macrophages, and dendritic cells. Science 327:656–661
58. Lu H, Zhang C, Wu W et al (2023) MCPIP1 restrains mucosal inflammation by orchestrating the intestinal monocyte to macrophage maturation via an ATF3-AP1S2 axis. Gut 72:882–895

59. Kim DH, Kim HC, Im K et al (2025) Inhibition of AXL ameliorates pulmonary fibrosis via attenuation of M2 macrophage polarisation. Eur Respir J. https://doi.org/10.1183/13993003.00615-2024
60. Janssen H, Kahles F, Liu D et al (2023) Monocytes re-enter the bone marrow during fasting and alter the host response to infection. Immunity 56:783-796.e787
61. Ruder AV, Wetzels SMW, Temmerman L, Biessen EAL, Goossens P (2023) Monocyte heterogeneity in cardiovascular disease. Cardiovasc Res 119:2033–2045
62. Selimoglu-Buet D, Wagner-Ballon O, Saada V et al (2015) Characteristic repartition of monocyte subsets as a diagnostic signature of chronic myelomonocytic leukemia. Blood 125:3618–3626
63. Guilliams M, Mildner A, Yona S (2018) Developmental and functional heterogeneity of monocytes. Immunity 49:595–613
64. Shi C, Pamer EG (2011) Monocyte recruitment during infection and inflammation. Nat Rev Immunol 11:762–774
65. Huang X, Yao Y, Hou X et al (2022) Macrophage SCAP contributes to metaflammation and lean NAFLD by activating STING-NF-κB signaling pathway. Cell Mol Gastroenterol Hepatol 14:1–26
66. Zhuang L, Zong X, Yang Q, Fan Q, Tao R (2023) Interleukin-34-NF-κB signaling aggravates myocardial ischemic/reperfusion injury by facilitating macrophage recruitment and polarization. EBioMedicine 95:104744
67. Zhang L, Fang F, Liu D et al (2025) Early secretory antigen target of 6-kDa of *Mycobacterium tuberculosis* inhibits macrophage apoptosis and host defense via TLR2. Respir Res 26:131
68. Abuaita BH, Burkholder KM, Boles BR, O'Riordan MX (2015) The endoplasmic reticulum stress sensor inositol-requiring enzyme 1α augments bacterial killing through sustained oxidant production. MBio 6:e00705
69. Ma C, Wang C, Zhang Y et al (2023) Phillygenin inhibited M1 macrophage polarization and reduced hepatic stellate cell activation by inhibiting macrophage exosomal miR-125b-5p. Biomed Pharmacother 159:114264
70. Li B, Xia C, He W et al (2024) The thyroid hormone analog GC-1 mitigates acute lung injury by inhibiting M1 macrophage polarization. Adv Sci 11:e2401931
71. Rui Y, Han X, Jiang A et al (2022) Eucalyptol prevents bleomycin-induced pulmonary fibrosis and M2 macrophage polarization. Eur J Pharmacol 931:175184
72. Yang X, Liu Z, Zhou J et al (2024) SPP1 promotes the polarization of M2 macrophages through the Jak2/Stat3 signaling pathway and accelerates the progression of idiopathic pulmonary fibrosis. Int J Mol Med 54(4):89
73. Wang Y, Zhang L, Wu GR et al (2021) MBD2 serves as a viable target against pulmonary fibrosis by inhibiting macrophage M2 program. Sci Adv. https://doi.org/10.1126/sciadv.abb6075
74. Zhang R, Jiang Q, Gao S et al (2024) Favipiravir ameliorates bleomycin-induced pulmonary fibrosis by reprogramming M1/M2 macrophage polarization. Int Immunopharmacol 131:111774
75. Xu H, Ma K, Ma Z, Zhuang T, Lin L (2025) Iguratimod improves bleomycin-induced pulmonary inflammation and fibrosis by regulating macrophage polarization through inhibiting the TLR4/NF-κB pathway. Front Immunol 16:1558903
76. Li X, Mara AB, Musial SC et al (2024) Coordinated chemokine expression defines macrophage subsets across tissues. Nat Immunol 25:1110–1122
77. Ural BB, Yeung ST, Damani-Yokota P et al (2020) Identification of a nerve-associated, lung-resident interstitial macrophage subset with distinct localization and immunoregulatory properties. Sci Immunol 5(45):eaax8756
78. Abdulnour RE, Sham HP, Douda DN et al (2016) Aspirin-triggered resolvin D1 is produced during self-resolving gram-negative bacterial pneumonia and regulates host immune responses for the resolution of lung inflammation. Mucosal Immunol 9:1278–1287
79. Lee J, Mohammad N, Han K et al (2025) Alpha-defensins increase NTHi binding but not engulfment by the macrophages enhancing airway inflammation in Alpha-1 antitrypsin deficiency. Front Immunol 16:1543729
80. Pervizaj-Oruqaj L, Selvakumar B, Ferrero MR et al (2024) Alveolar macrophage-expressed Plet1 is a driver of lung epithelial repair after viral pneumonia. Nat Commun 15:87
81. Toussaint M, Fievez L, Drion PV et al (2013) Myeloid hypoxia-inducible factor 1α prevents airway allergy in mice through macrophage-mediated immunoregulation. Mucosal Immunol 6:485–497
82. Zhu S (2021) Monocyte as a prognostic marker in patients with idiopathic pulmonary fibrosis. Respir Res 22:270
83. Kaul B, Petersen LA, Collard HR et al (2025) Prognostic Significance of Peripheral Blood Monocyte Count Among a National Cohort of Veterans with Pulmonary Fibrosis. Ann Am Thoracic Soc. https://doi.org/10.1513/AnnalsATS.202410-1108OC
84. Kim HS, Ullevig SL, Zamora D, Lee CF, Asmis R (2012) Redox regulation of MAPK phosphatase 1 controls monocyte migration and macrophage recruitment. Proc Natl Acad Sci U S A 109:E2803-2812
85. Davis FM, Tsoi LC, Melvin WJ et al (2021) Inhibition of macrophage histone demethylase JMJD3 protects against abdominal aortic aneurysms. J Exp Med 218(6):e20201839
86. Cai D, Sun C, Murashita T, Que X, Chen SY (2023) ADAR1 non-editing function in macrophage activation and abdominal aortic aneurysm. Circ Res 132:e78–e93
87. Pei L, Hou Y, Feng Y et al (2023) Equine β-defensin 1 regulates cytokine expression and phagocytosis in *S. aureus*-infected mouse monocyte macrophages via the Paxillin-FAK-PI3K pathway. Int Immunopharmacol 123:110793
88. Zhang X, Zheng P, Prestwood TR et al (2020) Human regulatory dendritic cells develop from monocytes in response to signals from regulatory and helper T cells. Front Immunol 11:1982
89. Dadey RE, Li R, Griner J et al (2025) Multiomics identifies tumor-intrinsic SREBP1 driving immune exclusion in hepatocellular carcinoma. J Immunother Cancer. https://doi.org/10.1136/jitc-2025-011537
90. Liu X, Zhang P, Zhang Y et al (2019) Glycolipid iGb3 feedback amplifies innate immune responses via CD1d reverse signaling. Cell Res 29:42–53
91. Kiemle-Kallee J, Kreipe H, Radzun HJ et al (1991) Alveolar macrophages in idiopathic pulmonary fibrosis display a more monocyte-like immunophenotype and an increased release of free oxygen radicals. Eur Respir J 4:400–406
92. Cheng D, Lian W, Wang T et al (2024) The interplay of Cxcl10(+)/Mmp14(+) monocytes and Ccl3(+) neutrophils proactively mediates silica-induced pulmonary fibrosis. J Hazard Mater 467:133713
93. Unterman A, Zhao AY, Neumark N et al (2024) Single-cell profiling reveals immune aberrations in progressive idiopathic pulmonary fibrosis. Am J Respir Crit Care Med 210:484–496
94. Poole JA, Schwab A, Thiele GM et al (2024) Unique transcriptomic profile of peripheral blood monocytes in rheumatoid arthritis-associated interstitial lung disease. Rheumatology (Oxford). https://doi.org/10.1093/rheumatology/keae572
95. Bailey JI, Puritz CH, Senkow KJ et al (2024) Profibrotic monocyte-derived alveolar macrophages are expanded in patients with persistent respiratory symptoms and radiographic abnormalities after COVID-19. Nat Immunol 25:2097–2109
96. Unterman A, Sumida TS, Nouri N et al (2022) Single-cell multiomics reveals dyssynchrony of the innate and adaptive immune system in progressive COVID-19. Nat Commun 13:440

97. Tourkina E, Richard M, Oates J et al (2010) Caveolin-1 regulates leucocyte behaviour in fibrotic lung disease. Ann Rheum Dis 69:1220–1226
98. Affandi AJ, Carvalheiro T, Ottria A et al (2022) CXCL4 drives fibrosis by promoting several key cellular and molecular processes. Cell Rep 38:110189
99. Liu SS, Liu C, Lv XX et al (2021) The chemokine CCL1 triggers an AMFR-SPRY1 pathway that promotes differentiation of lung fibroblasts into myofibroblasts and drives pulmonary fibrosis. Immunity 54:2042-2056.e2048
100. Fabre T, Barron AMS, Christensen SM et al (2023) Identification of a broadly fibrogenic macrophage subset induced by type 3 inflammation. Sci Immunol 8:eadd8945
101. Liu H, Fang S, Wang W et al (2016) Macrophage-derived MCPIP1 mediates silica-induced pulmonary fibrosis via autophagy. Part Fibre Toxicol 13:55
102. O'Dwyer DN, Armstrong ME, Kooblall M, Donnelly SC (2015) Targeting defective toll-like receptor-3 function and idiopathic pulmonary fibrosis. Expert Opin Ther Targets 19:507–514
103. Young LR, Borchers MT, Allen HL, Gibbons RS, McCormack FX (2006) Lung-restricted macrophage activation in the pearl mouse model of Hermansky-Pudlak syndrome. J Invest Med 176:4361–4368
104. Trujillo G, Meneghin A, Flaherty KR et al (2010) TLR9 differentiates rapidly from slowly progressing forms of idiopathic pulmonary fibrosis. Sci Transl Med 2:57ra82
105. Perrot CY, Karampitsakos T, Unterman A et al (2023) Mast-Cell Expressed Membrane Protein-1 (MCEMP1) is expressed in classical monocytes and alveolar macrophages in Idiopathic Pulmonary Fibrosis and regulates cell chemotaxis, adhesion, and migration in a TGFβ dependent manner. Am J Physiol Cell Physiol 326(3):C964–C977
106. Watanabe S, Markov NS, Lu Z et al (2021) Resetting proteostasis with ISRIB promotes epithelial differentiation to attenuate pulmonary fibrosis. Proc Natl Acad Sci U S A. https://doi.org/10.1073/pnas.2101100118
107. Zhao W, Wang L, Yang J et al (2023) Endothelial cell-derived MMP19 promotes pulmonary fibrosis by inducing E(nd)MT and monocyte infiltration. Cell Commun Signal 21:56
108. McCubbrey AL, Barthel L, Mohning MP et al (2018) Deletion of c-FLIP from CD11b(hi) macrophages prevents development of bleomycin-induced lung fibrosis. Am J Respir Cell Mol Biol 58:66–78
109. Cheresh P, Kim SJ, Jablonski R et al (2021) SIRT3 overexpression ameliorates asbestos-induced pulmonary fibrosis, mt-DNA damage, and lung fibrogenic monocyte recruitment. Int J Mol Sci. https://doi.org/10.3390/ijms22136856
110. Farhat A, Radhouani M, Deckert F et al (2025) An aging bone marrow exacerbates lung fibrosis by fueling profibrotic macrophage persistence. Sci Immunol 10:eadk5041
111. Hult EM, Gurczynski SJ, O'Dwyer DN et al (2022) Myeloid- and epithelial-derived heparin-binding epidermal growth factor-like growth factor promotes pulmonary fibrosis. Am J Respir Cell Mol Biol 67:641–653
112. Singh A, Chakraborty S, Wong SW et al (2022) Nanoparticle targeting of de novo profibrotic macrophages mitigates lung fibrosis. Proc Natl Acad Sci U S A 119:e2121098119
113. Wang S, Li J, Wu C et al (2024) Single-cell RNA sequencing reveals monocyte-derived interstitial macrophages with a profibrotic phenotype in bleomycin-induced pulmonary fibrosis. Int J Mol Sci. https://doi.org/10.3390/ijms252111669
114. Mia MM, Ghani S, Cibi DM et al (2025) YAP/TAZ are crucial regulators of macrophage-mediated pulmonary inflammation and fibrosis after bleomycin-induced injury. Eur Respir J. https://doi.org/10.1183/13993003.01544-2023
115. To S, Chavula T, Pedroza M, Smith J, Agarwal SK (2022) Cadherin-11 regulates macrophage development and function. Front Immunol 13:795337
116. Cui H, Jiang D, Banerjee S et al (2020) Monocyte-derived alveolar macrophage apolipoprotein E participates in pulmonary fibrosis resolution. JCI Insight 5(5):e134539
117. Zhang N, Yang K, Bai J et al (2020) Myeloid-specific blockade of Notch signaling alleviates murine pulmonary fibrosis through regulating monocyte-derived Ly6c(lo) MHCII(hi) alveolar macrophages recruitment and TGF-β secretion. FASEB J 34:11168–11184
118. Hong SY, Lu YT, Chen SY et al (2023) Targeting pathogenic macrophages by the application of SHP-1 agonists reduces inflammation and alleviates pulmonary fibrosis. Cell Death Dis 14:352
119. Cui H, Banerjee S, Xie N et al (2025) TREM2 promotes lung fibrosis via controlling alveolar macrophage survival and profibrotic activity. Nat Commun 16:1761
120. Cheng PP, He XL, Jia ZH et al (2025) Midkine, a novel MCP-1 activator mediated PM2.5-aggravated experimental pulmonary fibrosis. Environ Int 197:109354
121. Bu N, Gao Y, Zhao Y et al (2022) LncRNA H19 via miR-29a-3p is involved in lung inflammation and pulmonary fibrosis induced by neodymium oxide. Ecotoxicol Environ Saf 247:114173
122. Pilling D, Vakil V, Cox N, Gomer RH (2015) TNF-α-stimulated fibroblasts secrete lumican to promote fibrocyte differentiation. Proc Natl Acad Sci U S A 112:11929–11934
123. Liu B, Jiang Q, Chen R et al (2024) Tacrolimus alleviates pulmonary fibrosis progression through inhibiting the activation and interaction of ILC2 and monocytes. Int Immunopharmacol 132:111999
124. Maksimova A, Shevela E, Sakhno L, Tikhonova M, Ostanin A, Chernykh E (2023) Human macrophages polarized by interaction with apoptotic cells produce fibrosis-associated mediators and enhance pro-fibrotic activity of dermal fibroblasts in vitro. Cells. https://doi.org/10.3390/cells12151928
125. Kwak D, Bradley PB, Subbotina N et al (2023) CD36/Lyn kinase interactions within macrophages promotes pulmonary fibrosis in response to oxidized phospholipid. Respir Res 24:314
126. Kwak D, Ling S, Subbotina N, Shayman JA, Sisson TH, Kim KK (2025) Degradation of oxidized phospholipids by lysosomal phospholipase A2 regulates pulmonary fibrosis. Am J Physiol Lung Cell Mol Physiol 329:L214-l223
127. Fukushima K, Akira S (2021) Novel insights into the pathogenesis of lung fibrosis: the RBM7-NEAT1-CXCL12-SatM axis at fibrosis onset. Int Immunol 33:659–663
128. Parajuli N, Yao Y, Khalasawi N et al (2025) MicroRNAs regulate alveolar macrophage homeostasis and its function in lung fibrosis. Front Immunol 16:1598306
129. Larson-Casey JL, Gu L, Kang J, Dhyani A, Carter AB (2021) Nox4 regulates macrophage apoptosis resistance to induce fibrotic progression. J Biol Chem 297:100810
130. Palomäki J, Sund J, Vippola M et al (2015) A secretomics analysis reveals major differences in the macrophage responses towards different types of carbon nanotubes. Nanotoxicology 9:719–728
131. Grunwell JR, Yeligar SM, Stephenson S et al (2018) TGF-β1 suppresses the type i IFN response and induces mitochondrial dysfunction in alveolar macrophages. J Immunol 200:2115–2128
132. Lemos DR, Babaeijandaghi F, Low M et al (2015) Nilotinib reduces muscle fibrosis in chronic muscle injury by promoting TNF-mediated apoptosis of fibro/adipogenic progenitors. Nat Med 21:786–794
133. Fu P, Han X, Lin QY et al (2025) CD11b Blockade Ameliorates Myocardial Ischemia/Reperfusion Injury by Reducing Neutrophil and Monocyte Infiltration. J Am Heart Assoc 14:e038142
134. Zhao K, Yang T, Sun M et al (2017) IP-10 expression in patients with chronic HBV infection and its ability to predict the decrease

in HBsAg levels after treatment with Entecavir. Molecules Cells 40:418–425
135. Alsina-Sanchis E, Mülfarth R, Moll I, Mogler C, Rodriguez-Vita J, Fischer A (2021) Intraperitoneal oil application causes local inflammation with depletion of resident peritoneal macrophages. Mol Cancer Res 19:288–300
136. Bellamri N, Morzadec C, Joannes A et al (2019) Alteration of human macrophage phenotypes by the anti-fibrotic drug nintedanib. Int Immunopharmacol 72:112–123
137. Moore ZR, Huang X, Lobaugh S et al (2024) Biomarkers associated with pulmonary exacerbations in a randomized trial of nintedanib for radiation pneumonitis. Radiother Oncol 196:110320
138. Pan L, Cheng Y, Yang W et al (2023) Nintedanib Ameliorates Bleomycin-Induced Pulmonary Fibrosis, Inflammation, Apoptosis, and Oxidative Stress by Modulating PI3K/Akt/mTOR Pathway in Mice. Inflammation 46:1531–1542
139. Zhang G, Shi L, Li J et al (2023) Antler stem cell exosomes alleviate pulmonary fibrosis via inhibiting recruitment of monocyte macrophage, rather than polarization of M2 macrophages in mice. Cell Death Discov 9:359
140. Choi SM, Mo Y, Bang JY et al (2023) Classical monocyte-derived macrophages as therapeutic targets of umbilical cord mesenchymal stem cells: comparison of intratracheal and intravenous administration in a mouse model of pulmonary fibrosis. Respir Res 24:68
141. Mansouri N, Willis GR, Fernandez-Gonzalez A et al (2019) Mesenchymal stromal cell exosomes prevent and revert experimental pulmonary fibrosis through modulation of monocyte phenotypes. JCI Insight 4(21):e128060
142. Miao YM, Zhang YJ, Qiao SM, Xia YF, Wei ZF, Dai Y (2021) Oral administration of curcumin ameliorates pulmonary fibrosis in mice through 15d-PGJ2-mediated induction of hepatocyte growth factor in the colon. Acta Pharmacol Sin 42:422–435
143. Okuma T, Terasaki Y, Kaikita K et al (2004) C-C chemokine receptor 2 (CCR2) deficiency improves bleomycin-induced pulmonary fibrosis by attenuation of both macrophage infiltration and production of macrophage-derived matrix metalloproteinases. J Pathol 204:594–604
144. Shichino S, Abe J, Ueha S et al (2015) Reduced supply of monocyte-derived macrophages leads to a transition from nodular to diffuse lesions and tissue cell activation in silica-induced pulmonary fibrosis in mice. Am J Pathol 185:2923–2938
145. Joshi N, Watanabe S, Verma R et al (2020) A spatially restricted fibrotic niche in pulmonary fibrosis is sustained by M-CSF/M-CSFR signalling in monocyte-derived alveolar macrophages. Eur Respir J. https://doi.org/10.1183/13993003.00646-2019
146. Venosa A, Cowman S, Katzen J et al (2021) Role of CCR2(+) myeloid cells in inflammation responses driven by expression of a surfactant protein-C mutant in the alveolar epithelium. Front Immunol 12:665818
147. Brody SL, Gunsten SP, Luehmann HP et al (2021) Chemokine receptor 2-targeted molecular imaging in pulmonary fibrosis. A clinical trial. Am J Respir Crit Care Med 203:78–89
148. Chen B, Yang Y, Yang C et al (2021) M2 macrophage accumulation contributes to pulmonary fibrosis, vascular dilatation, and hypoxemia in rat hepatopulmonary syndrome. J Cell Physiol 236:7682–7697
149. Farooq H, Luehmann HP, Koenitzer JR et al (2024) Molecular imaging in experimental pulmonary fibrosis reveals that nintedanib unexpectedly modulates CCR2 immune cell infiltration. EBioMedicine 110:105431
150. Moos PJ, Cheminant JR, Cowman S et al (2024) Spatial and phenotypic heterogeneity of resident and monocyte-derived macrophages during inflammatory exacerbations leading to pulmonary fibrosis. Front Immunol 15:1425466

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